

# Informe de Validación de Scorecard

USO INTERNO – CONFIDENCIAL

|                            |                     |                  |
|----------------------------|---------------------|------------------|
| MODELO                     | ENTIDAD             | CARTERA          |
|                            |                     |                  |
| RESPONSABLE DEL DESARROLLO | VERSIÓN DEL INFORME | FECHA DE EMISIÓN |
|                            |                     | 2026-07-21       |

TRAZABILIDAD DE LA CORRIDA (LINEAGE)

config\_hash=4b93809debd102cf1644f8b5acaa9abbd693204e9f3fb7346f392425677ae49f  
data\_hash=74c5d9ddeca8ff71853623d20ace015d0788564549339d61bfb66f24dfadfcc8  
git\_sha=ad5fe58dc5eededee97a4d4d7c3c4b8ef979af  
root\_seed=20240706

|                |                          |                |
|----------------|--------------------------|----------------|
| DESARROLLO     | VALIDACIÓN INDEPENDIENTE | APROBACIÓN     |
|                |                          |                |
| Nombre y fecha | Nombre y fecha           | Nombre y fecha |

## Resumen ejecutivo

### Veredicto de validación

☐ Aprobado    ☐ Aprobado con observaciones    ☐ Rechazado

Campo estructural: lo marca y firma el validador independiente; no es un resultado calculado por el motor.

| Métrica                             | Alcance                 | Valor                 | Estado                                                                                              |
|-------------------------------------|-------------------------|-----------------------|-----------------------------------------------------------------------------------------------------|
| AUC                                 | Desarrollo              | 0,7123                |  Sin alertas     |
| Gini                                | Desarrollo              | 0,4247                |  Sin alertas     |
| KS                                  | Desarrollo              | 0,3201                |  Sin alertas     |
| AUC                                 | Holdout                 | 0,6946                |  Sin alertas     |
| Gini                                | Holdout                 | 0,3893                |  Sin alertas     |
| KS                                  | Holdout                 | 0,3119                |  Sin alertas   |
| AUC                                 | Fuera de tiempo (OOT)   | 0,6561                |  Sin alertas   |
| Gini                                | Fuera de tiempo (OOT)   | 0,3122                |  Sin alertas   |
| KS                                  | Fuera de tiempo (OOT)   | 0,2519                |  Sin alertas   |
| PSI del score                       | Desarrollo vs. Holdout  | 0,0132                |  Estable       |
| PSI del score                       | Desarrollo vs. OOT      | 0,0068                |  Estable       |
| Estado técnico de validación formal | preset-estandar-consumo | 3 de 3 tests fallidos |  Falla técnica |

No se configuraron umbrales de discriminación, de modo que el motor no puede levantar alertas sobre AUC, Gini ni KS: «sin alertas» significa que no había umbral contra el cual comparar, no que la métrica sea satisfactoria.

Las bandas de PSI usan los umbrales configurados: estable por debajo de 0,10 y redesarrollo por sobre 0,25.

La validación formal ejecutó discriminación, calibración y estabilidad. Su estado es una síntesis técnica del motor y no sustituye el veredicto que debe firmar un validador humano.

Alcance: scorecard y validación formal de las familias configuradas. El capítulo «Validación formal» reproduce el resultado técnico atómico; el veredicto permanece

bajo responsabilidad humana. Todos los valores provienen de la corrida trazada en la portada: no se completan con supuestos.

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## 1 Introducción

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### **POR COMPLETAR — INTRODUCCIÓN**

Declara el propósito del informe y qué decisión habilita: ¿es una validación inicial previa a la puesta en producción, una revisión periódica o un recalibrado?

Delimita el alcance (qué modelo y qué cartera cubre, qué queda fuera) e identifica la audiencia: Validación independiente, Comité de Riesgo, regulador.

## 2 Contexto del modelo y de la cartera

La población procesada contiene 6.000 observaciones, 8 variables de entrada y una tasa de incumplimiento de 23,45 %. Los conteos por estado, partición y motivo de exclusión se reproducen literalmente en las tablas de esta sección.

La columna objetivo publicada por data es «target».

La variable objetivo del ejercicio es «target», construida por las reglas de incumplimiento declaradas en el config (Anexo C). La definición de negocio que justifica esas reglas —p. ej. 90+ días de mora— debe explicitarla el analista: el motor aplica la regla, no la fundamenta.

La población se particionó por cohorte sobre «cohorte», reservando como OOT la cohorte «2024Q2», y un Holdout de 20,00 % del resto.

### POR COMPLETAR — CONTEXTO DEL MODELO

Describe la cartera sobre la que se construyó el modelo: producto, segmento, volumen y período.

Explicita la definición de incumplimiento utilizada (p. ej. 90+ días de mora) y por qué; declara las exclusiones aplicadas a la población y su justificación.

### 2.1 Población, particiones y exclusiones

Las tablas siguientes son una proyección de presentación del DataCardSection publicado por data. Estados, tamaños y tasas por partición, y exclusiones se copian literalmente; el informe no recalcula ni completa valores ausentes.

#### ESTADOS DE LA POBLACIÓN

| Estado        | Observaciones |
|---------------|---------------|
| bueno         | 4593          |
| malo          | 1407          |
| indeterminado | 0             |
| excluido      | 0             |

#### PARTICIONES DE MODELAMIENTO

| Partición  | Observaciones | Tasa de incumplimiento |
|------------|---------------|------------------------|
| desarrollo | 3961          | 0.2333                 |
| holdout    | 1031          | 0.2367                 |

| Partición       | Observaciones | Tasa de incumplimiento |
|-----------------|---------------|------------------------|
| oot             | 1008          | 0.2371                 |
| fuera_de_modelo | 0             | 0.0000                 |

**EXCLUSIONES APLICADAS**

| Motivo | Exclusiones |
|--------|-------------|
|--------|-------------|



## 3 Metodología

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Este capítulo describe el procedimiento realmente ejecutado por el motor en esta corrida, con los parámetros efectivos que constan en el config trazado en el Anexo A. No es una descripción genérica de la técnica: cada cifra citada aquí es la que produjo este modelo.

### 3.1 Tratamiento de datos y particiones

El dataset se validó contra un esquema declarado de 8 columnas, con tipos y nulabilidad explícitos; una desviación del esquema detiene la corrida en vez de propagarse.

En el tratamiento de faltantes se rechaza toda variable con más de 99,00 % de valores faltantes.

### 3.2 Binning WoE

Se discretizó cada variable con OptBinning, resolviendo el problema óptimo por programación entera mixta (MIP), partiendo de un máximo de 20 pre-bins, con un máximo de 6 bins por variable y un tamaño mínimo de bin de 5,00 % de la población.

La tendencia monótona exigida al WoE es automática (ascendente o descendente): el riesgo debe comportarse de forma coherente entre tramos contiguos, sin quiebres que un analista no pueda defender.

De 6 variables candidatas, 6 quedaron binificadas.

Los valores especiales se agrupan en un bin propio y los faltantes reciben el tratamiento «empirical» (OptBinning 0.20.0).

### 3.3 Selección de variables

Se descartaron las variables con IV inferior a 0,02; las de IV superior a 0,50 se marcaron para revisión, porque un poder predictivo tan alto suele indicar fuga de información más que señal legítima.

Se eliminó la redundancia entre pares de variables con correlación (pearson) sobre 0,75 y se acotó la multicolinealidad exigiendo un VIF bajo 5,0.

El filtro dejó 5 de 6 variables candidatas; el motivo de exclusión de cada variable descartada consta en la tabla de selección.

### 3.4 Estimación del modelo PD

La probabilidad de incumplimiento se modeló mediante regresión logística (logit) sobre las variables transformadas a WoE, con un nivel de significancia de 0,05.

La construcción del modelo usó stepwise bidireccional (criterio «wald\_pvalue» y p-valor de entrada 0,05 y de salida 0,05).

Se exige que el coeficiente de cada variable en WoE tenga el signo esperado (negativo): cuando se invierte, se marca la variable sin excluirla. Un signo invertido significa que la variable contradice la relación de riesgo que su binning declara.

El modelo final retiene 5 de 5 variables de entrada.

### 3.5 Escalado del scorecard

El modelo se escaló a puntaje con la convención PDO: un puntaje de 600 corresponde a odds de 50:1, y cada 20 puntos adicionales duplican esas odds.

De esa convención se derivan los parámetros efectivos de la escala: factor 28,8539 y offset 487,1229.

Un puntaje más alto indica menor riesgo y los puntajes se emiten con redondeo al entero más cercano.

### 3.6 Calibración de la PD

La PD se calibró por desplazamiento del intercepto (intercept offset), ajustando sobre la partición de Desarrollo contra un ancla de negocio declarada explícitamente, con una PD objetivo de 20,00 %. El ancla es a través del ciclo (TTC).

El desplazamiento aplicado al intercepto es -0,2184, lo que preserva el ordenamiento del score: la calibración corrige el nivel de la PD, no su ranking.

El ajuste se realizó sobre 3961 observaciones de Desarrollo.

## 4 Resultados

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Los resultados se presentan por etapa, con las tablas que sostienen el juicio de validación. El detalle exhaustivo —todas las tablas y todos los parámetros de cada paso— no se pierde: vive en los Anexos B y C, que son parte de este mismo informe.

### 4.1 Binning WoE y poder predictivo

El poder predictivo univariado (IV) de las 6 variables binificadas lo encabezan «ingreso\_mensual» (0,305), «deuda\_ingreso» (0,164) y «utilizacion\_linea» (0,061). El IV de cada variable consta en la tabla siguiente y su binning completo, bin a bin, en el Anexo B.

La monotonía efectiva del WoE resultó en 3 variables con tendencia ascendente y 2 variables con tendencia descendente, y 1 sin tendencia monótona resuelta, lo que debe revisarse.

## RESUMEN DE BINNING — IV POR VARIABLE

| name              | dtype       | status  | selected | n_bins | iv       | js       | gini     | quality_score | iv_band | is_suspicious_iv | is_zero_iv | monotonic_trend | skipped_reason |
|-------------------|-------------|---------|----------|--------|----------|----------|----------|---------------|---------|------------------|------------|-----------------|----------------|
| ingreso_mensual   | numerical   | OPTIMAL | true     | 6      | 0.304750 | 0.036713 | 0.294575 | 0.818100      | strong  | false            | false      | descending      | —              |
| deuda_ingreso     | numerical   | OPTIMAL | true     | 6      | 0.163519 | 0.019897 | 0.204769 | 0.171203      | medium  | false            | false      | ascending       | —              |
| utilizacion_linea | numerical   | OPTIMAL | true     | 5      | 0.061031 | 0.007565 | 0.131539 | 0.173165      | weak    | false            | false      | ascending       | —              |
| mora_max_12m      | numerical   | OPTIMAL | true     | 6      | 0.021870 | 0.002727 | 0.079733 | 0.003140      | weak    | false            | false      | ascending       | —              |
| antiguedad_meses  | numerical   | OPTIMAL | true     | 6      | 0.041379 | 0.005099 | 0.095492 | 0.005044      | weak    | false            | false      | descending      | —              |
| segmento          | categorical | OPTIMAL | true     | 3      | 0.002922 | 0.000365 | 0.029244 | 0.002656      | —       | false            | false      | —               | —              |

## 4.2 Selección de variables

Las 5 variables que pasaron el filtro son «antiguedad\_meses», «deuda\_ingreso», «ingreso\_mensual», «mora\_max\_12m» y «utilizacion\_linea».

Las exclusiones se reparten en 1 por IV insuficiente.

Tras la selección, la correlación absoluta máxima entre las variables finales es 0,0303 y el VIF máximo es 1,00.

## CRITERIOS DE SELECCIÓN POR VARIABLE

| feature           | woe_column             | included | reason   | iv       | iv_band | auc      | gini     | ks       | max_abs_corr | max_corr_with | vif      | max_csi | forced | detail                            |
|-------------------|------------------------|----------|----------|----------|---------|----------|----------|----------|--------------|---------------|----------|---------|--------|-----------------------------------|
| antiguedad_meses  | antiguedad_meses__woe  | true     | included | 0.041379 | weak    | 0.547746 | 0.095492 | 0.060951 | —            | —             | 1.001115 | —       | —      | —                                 |
| deuda_ingreso     | deuda_ingreso__woe     | true     | included | 0.163519 | medium  | 0.602385 | 0.204769 | 0.160053 | —            | —             | 1.000130 | —       | —      | —                                 |
| ingreso_mensual   | ingreso_mensual__woe   | true     | included | 0.304750 | strong  | 0.647287 | 0.294575 | 0.211815 | —            | —             | 1.000568 | —       | —      | —                                 |
| mora_max_12m      | mora_max_12m__woe      | true     | included | 0.021870 | weak    | 0.539867 | 0.079733 | 0.064124 | —            | —             | 1.001595 | —       | —      | —                                 |
| segmento          | segmento__woe          | false    | low_iv   | 0.002922 | —       | 0.514622 | 0.029244 | 0.023399 | —            | —             | —        | —       | —      | iv=0.00292240935993 < min_iv=0.02 |
| utilizacion_linea | utilizacion_linea__woe | true     | included | 0.061031 | weak    | 0.565769 | 0.131539 | 0.097760 | —            | —             | 1.000578 | —       | —      | —                                 |

## FACTOR DE INFLACIÓN DE VARIANZA (VIF)

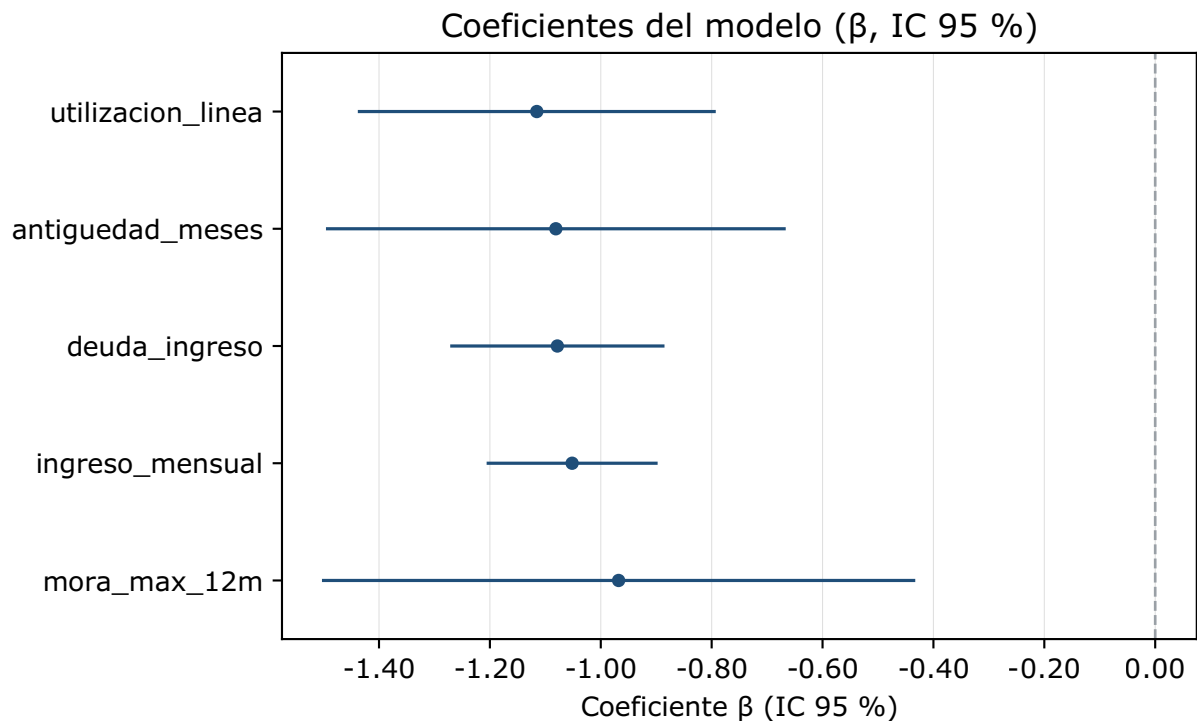
| iteration | feature           | woe_column             | vif      | removed | reason |
|-----------|-------------------|------------------------|----------|---------|--------|
| 0         | ingreso_mensual   | ingreso_mensual__woe   | 1.000568 | false   | —      |
| 0         | deuda_ingreso     | deuda_ingreso__woe     | 1.000130 | false   | —      |
| 0         | utilizacion_linea | utilizacion_linea__woe | 1.000578 | false   | —      |
| 0         | antiguedad_meses  | antiguedad_meses__woe  | 1.001115 | false   | —      |
| 0         | mora_max_12m      | mora_max_12m__woe      | 1.001595 | false   | —      |

## 4.3 Modelo PD

El modelo final incluye 5 variables: «antiguedad\_meses», «deuda\_ingreso», «ingreso\_mensual», «mora\_max\_12m» y «utilizacion\_linea».

En el ajuste in-sample, el optimizador convergió, el ajuste usó 3961 observaciones de Desarrollo, de las cuales 924 son incumplimientos y el pseudo-R<sup>2</sup> de McFadden es 0,0968.

Marcadas por concentrar una contribución excesiva al IV del modelo:  
«ingreso\_mensual».



## COEFICIENTES DEL MODELO PD

| feature           | woe_column             | beta      | standard_error | wald_z     | p_value  | conf_low  | conf_high | expected_sign | sign_ok | iv       | iv_contribution |
|-------------------|------------------------|-----------|----------------|------------|----------|-----------|-----------|---------------|---------|----------|-----------------|
| intercept         | const                  | -1.192119 | 0.040521       | -29.419899 | 0.000000 | -1.271538 | -1.112699 | —             | —       | —        | —               |
| antiguedad_meses  | antiguedad_meses__woe  | -1.081039 | 0.210069       | -5.146120  | 0.000000 | -1.492767 | -0.669312 | negative      | true    | 0.041379 | 0.069833        |
| deuda_ingreso     | deuda_ingreso__woe     | -1.078315 | 0.097124       | -11.102485 | 0.000000 | -1.268674 | -0.887956 | negative      | true    | 0.163519 | 0.275958        |
| ingreso_mensual   | ingreso_mensual__woe   | -1.051670 | 0.077207       | -13.621497 | 0.000000 | -1.202992 | -0.900348 | negative      | true    | 0.304750 | 0.514303        |
| mora_max_12m      | mora_max_12m__woe      | -0.967824 | 0.271521       | -3.564456  | 0.000365 | -1.499995 | -0.435653 | negative      | true    | 0.021870 | 0.036909        |
| utilizacion_linea | utilizacion_linea__woe | -1.115409 | 0.163226       | -6.833521  | 0.000000 | -1.435327 | -0.795492 | negative      | true    | 0.061031 | 0.102998        |

## 4.4 Scorecard

El scorecard asigna puntajes a los tramos de 5 variables.

## SCORECARD — PUNTAJES POR ATRIBUTO

| feature          | woe_column            | bin_label              | bin_index | woe       | beta      | intercept_share | raw_points | points | rounding_delta | source        |
|------------------|-----------------------|------------------------|-----------|-----------|-----------|-----------------|------------|--------|----------------|---------------|
| antiguedad_meses | antiguedad_meses__woe | (-inf, 12.50)          | 0         | -0.315182 | -1.081039 | -0.238424       | 94.472796  | 94     | -0.472796      | binning_table |
| antiguedad_meses | antiguedad_meses__woe | [12.50, 36.50)         | 1         | -0.134523 | -1.081039 | -0.238424       | 100.107959 | 100    | -0.107959      | binning_table |
| antiguedad_meses | antiguedad_meses__woe | [36.50, 43.50)         | 2         | -0.018542 | -1.081039 | -0.238424       | 103.725671 | 104    | 0.274329       | binning_table |
| antiguedad_meses | antiguedad_meses__woe | [43.50, 89.50)         | 3         | 0.005106  | -1.081039 | -0.238424       | 104.463290 | 104    | -0.463290      | binning_table |
| antiguedad_meses | antiguedad_meses__woe | [89.50, 114.50)        | 4         | 0.134868  | -1.081039 | -0.238424       | 108.510864 | 109    | 0.489136       | binning_table |
| antiguedad_meses | antiguedad_meses__woe | [114.50, inf)          | 5         | 0.750487  | -1.081039 | -0.238424       | 127.713374 | 128    | 0.286626       | binning_table |
| antiguedad_meses | antiguedad_meses__woe | Special                | 6         | 0.000000  | -1.081039 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |
| antiguedad_meses | antiguedad_meses__woe | Missing                | 7         | 0.000000  | -1.081039 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |
| deuda_ingreso    | deuda_ingreso__woe    | (-inf, 0.08)           | 0         | 0.713764  | -1.078315 | -0.238424       | 126.511783 | 127    | 0.488217       | binning_table |
| deuda_ingreso    | deuda_ingreso__woe    | [0.08, 0.16)           | 1         | 0.272018  | -1.078315 | -0.238424       | 112.767495 | 113    | 0.232505       | binning_table |
| deuda_ingreso    | deuda_ingreso__woe    | [0.16, 0.37)           | 2         | 0.217921  | -1.078315 | -0.238424       | 111.084347 | 111    | -0.084347      | binning_table |
| deuda_ingreso    | deuda_ingreso__woe    | [0.37, 0.51)           | 3         | -0.103496 | -1.078315 | -0.238424       | 101.083886 | 101    | -0.083886      | binning_table |
| deuda_ingreso    | deuda_ingreso__woe    | [0.51, 0.85)           | 4         | -0.390210 | -1.078315 | -0.238424       | 92.163191  | 92     | -0.163191      | binning_table |
| deuda_ingreso    | deuda_ingreso__woe    | [0.85, inf)            | 5         | -1.082493 | -1.078315 | -0.238424       | 70.623791  | 71     | 0.376209       | binning_table |
| deuda_ingreso    | deuda_ingreso__woe    | Special                | 6         | 0.000000  | -1.078315 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |
| deuda_ingreso    | deuda_ingreso__woe    | Missing                | 7         | 0.000000  | -1.078315 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |
| ingreso_mensual  | ingreso_mensual__woe  | (-inf, 242795.88)      | 0         | -0.902231 | -1.051670 | -0.238424       | 76.926018  | 77     | 0.073982       | binning_table |
| ingreso_mensual  | ingreso_mensual__woe  | [242795.88, 354256.50) | 1         | -0.582461 | -1.051670 | -0.238424       | 86.629380  | 87     | 0.370620       | binning_table |
| ingreso_mensual  | ingreso_mensual__woe  | [354256.50, 464805.09) | 2         | -0.253561 | -1.051670 | -0.238424       | 96.609779  | 97     | 0.390221       | binning_table |
| ingreso_mensual  | ingreso_mensual__woe  | [464805.09, 631639.47) | 3         | 0.036404  | -1.051670 | -0.238424       | 105.408702 | 105    | -0.408702      | binning_table |
| ingreso_mensual  | ingreso_mensual__woe  | [631639.47, 913196.97) | 4         | 0.427766  | -1.051670 | -0.238424       | 117.284504 | 117    | -0.284504      | binning_table |
| ingreso_mensual  | ingreso_mensual__woe  | [913196.97, inf)       | 5         | 1.209914  | -1.051670 | -0.238424       | 141.018611 | 141    | -0.018611      | binning_table |
| ingreso_mensual  | ingreso_mensual__woe  | Special                | 6         | 0.000000  | -1.051670 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |
| ingreso_mensual  | ingreso_mensual__woe  | Missing                | 7         | 0.000000  | -1.051670 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |
| mora_max_12m     | mora_max_12m__woe     | (-inf, 2.50)           | 0         | 0.360684  | -0.967824 | -0.238424       | 114.376309 | 114    | -0.376309      | binning_table |
| mora_max_12m     | mora_max_12m__woe     | [2.50, 3.50)           | 1         | 0.214081  | -0.967824 | -0.238424       | 110.282335 | 110    | -0.282335      | binning_table |
| mora_max_12m     | mora_max_12m__woe     | [3.50, 5.50)           | 2         | 0.091020  | -0.967824 | -0.238424       | 106.845820 | 107    | 0.154180       | binning_table |
| mora_max_12m     | mora_max_12m__woe     | [5.50, 7.50)           | 3         | -0.075227 | -0.967824 | -0.238424       | 102.203270 | 102    | -0.203270      | binning_table |
| mora_max_12m     | mora_max_12m__woe     | [7.50, 9.50)           | 4         | -0.126392 | -0.967824 | -0.238424       | 100.774458 | 101    | 0.225542       | binning_table |
| mora_max_12m     | mora_max_12m__woe     | [9.50, inf)            | 5         | -0.206456 | -0.967824 | -0.238424       | 98.538643  | 99     | 0.461357       | binning_table |
| mora_max_12m     | mora_max_12m__woe     | Special                | 6         | 0.000000  | -0.967824 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |

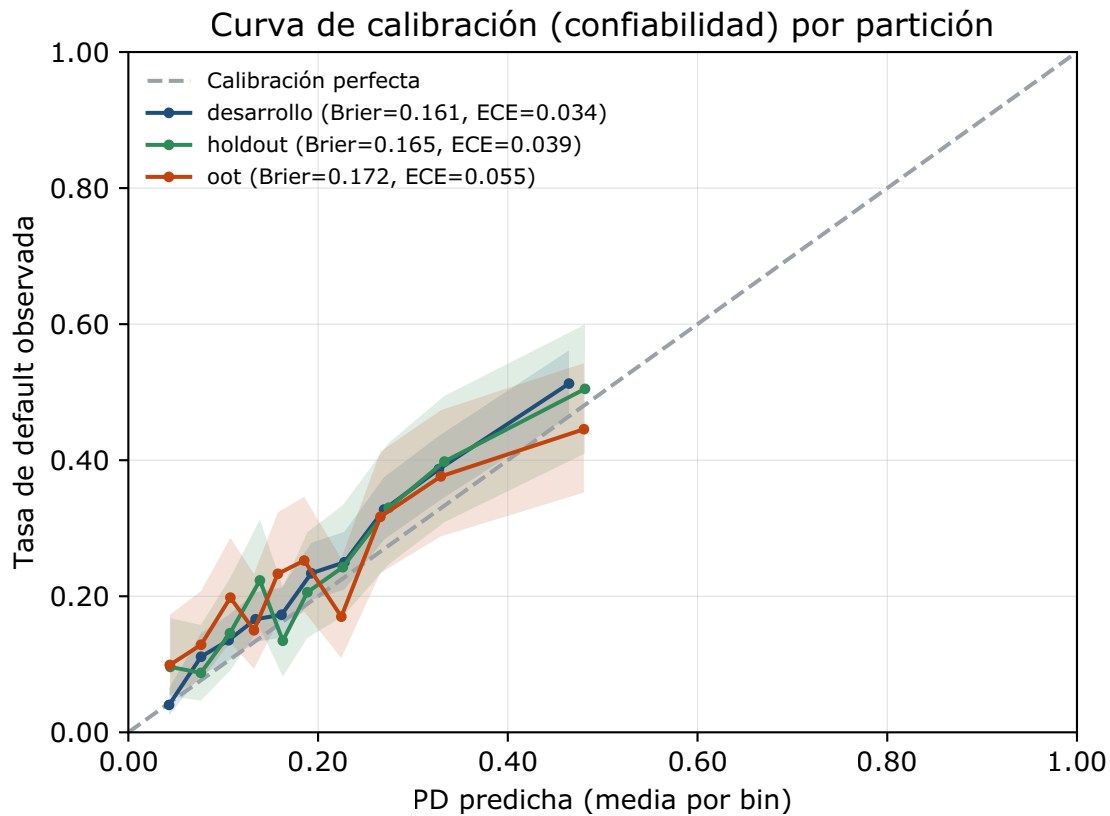


| feature           | woe_column             | bin_label    | bin_index | woe       | beta      | intercept_share | raw_points | points | rounding_delta | source        |
|-------------------|------------------------|--------------|-----------|-----------|-----------|-----------------|------------|--------|----------------|---------------|
| mora_max_12m      | mora_max_12m__woe      | Missing      | 7         | 0.000000  | -0.967824 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |
| utilizacion_linea | utilizacion_linea__woe | (-inf, 0.10) | 0         | 0.611896  | -1.115409 | -0.238424       | 123.997247 | 124    | 0.002753       | binning_table |
| utilizacion_linea | utilizacion_linea__woe | [0.10, 0.25) | 1         | 0.249576  | -1.115409 | -0.238424       | 112.336374 | 112    | -0.336374      | binning_table |
| utilizacion_linea | utilizacion_linea__woe | [0.25, 0.45) | 2         | 0.061922  | -1.115409 | -0.238424       | 106.296905 | 106    | -0.296905      | binning_table |
| utilizacion_linea | utilizacion_linea__woe | [0.45, 0.68) | 3         | -0.164768 | -1.115409 | -0.238424       | 99.001143  | 99     | -0.001143      | binning_table |
| utilizacion_linea | utilizacion_linea__woe | [0.68, inf)  | 4         | -0.465995 | -1.115409 | -0.238424       | 89.306503  | 89     | -0.306503      | binning_table |
| utilizacion_linea | utilizacion_linea__woe | Special      | 5         | 0.000000  | -1.115409 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |
| utilizacion_linea | utilizacion_linea__woe | Missing      | 6         | 0.000000  | -1.115409 | -0.238424       | 104.304029 | 104    | -0.304029      | binning_table |

## 4.5 Calibración

En Desarrollo, la PD media calibrada es 20,00 %, frente a 23,33 % antes de calibrar, contra una tasa de incumplimiento observada de 23,33 %.

La calibración preservó el ordenamiento de las observaciones: la capacidad discriminante del modelo no se altera.



## 4.6 Desempeño y discriminación

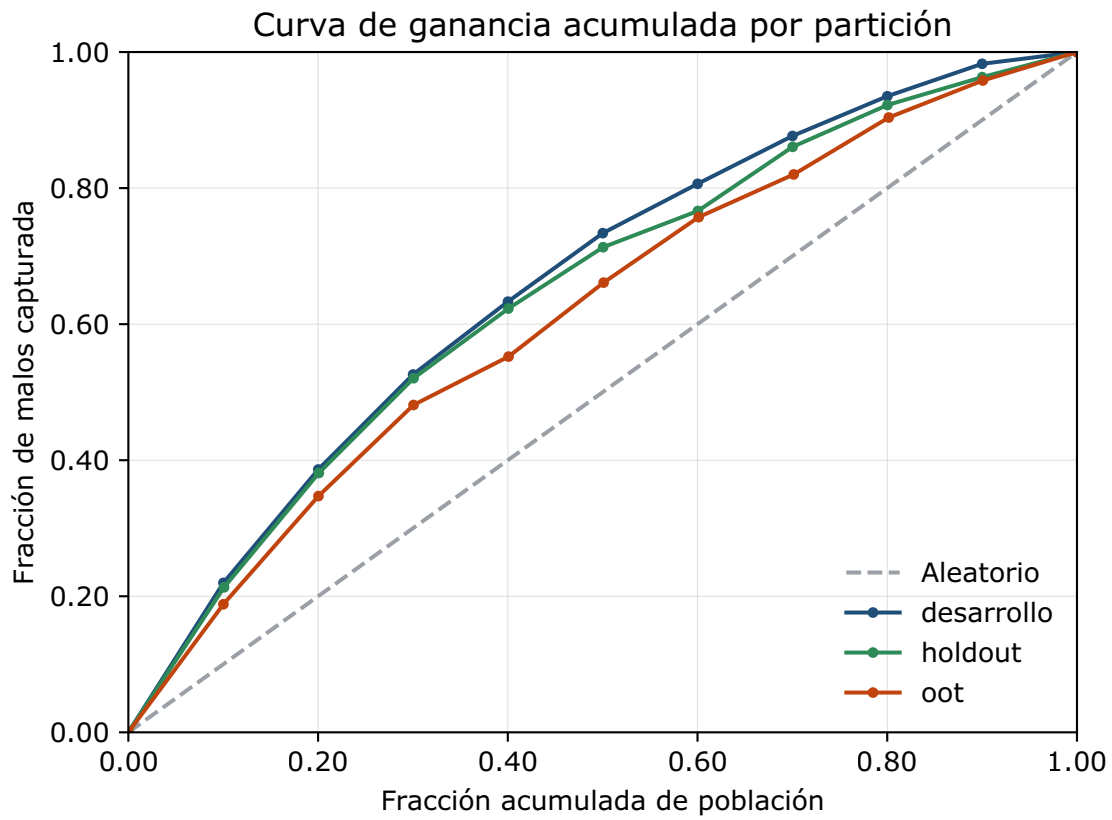
Las métricas de discriminación se calcularon sobre la PD calibrada.

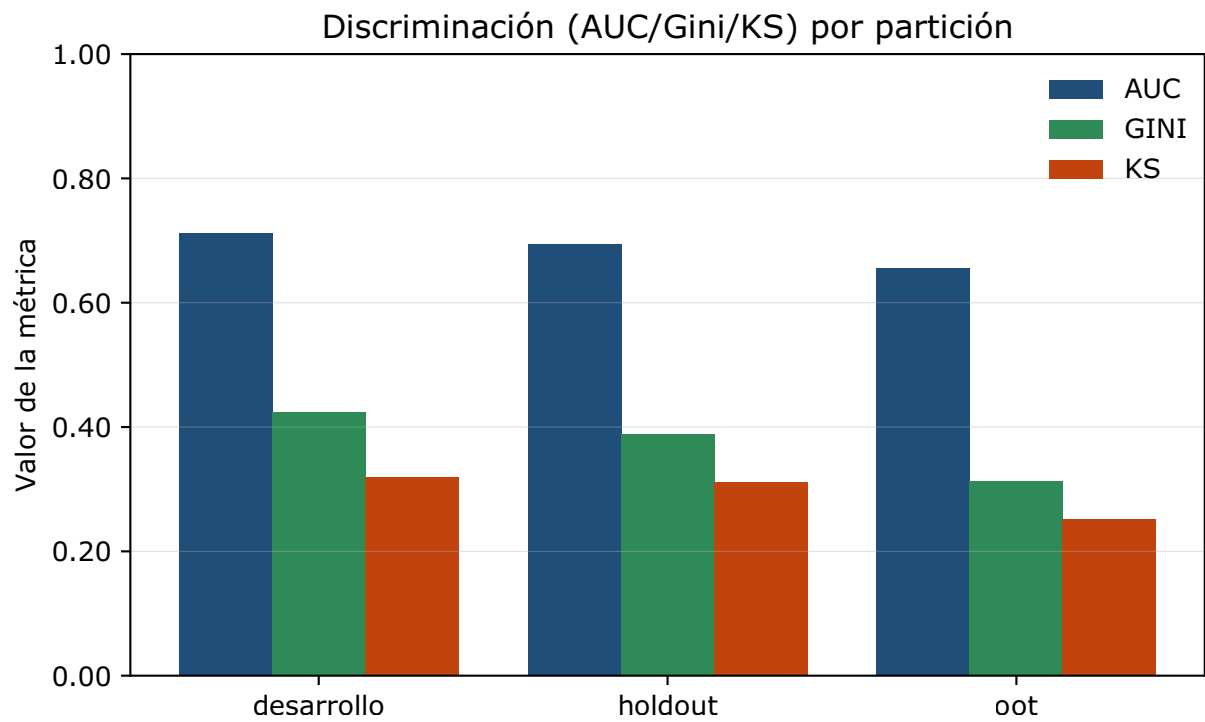
En Desarrollo el modelo alcanza AUC 0,7123, Gini 0,4247 y KS 0,3201.

En Holdout el modelo alcanza AUC 0,6946, Gini 0,3893 y KS 0,3119.

En Fuera de tiempo (OOT) el modelo alcanza AUC 0,6561, Gini 0,3122 y KS 0,2519.

La tabla de desempeño reparte la población en 10 tramos de riesgo ordenados por score, y permite verificar que la tasa de incumplimiento observada sea monótona entre tramos.





**MÉTRICAS DE DISCRIMINACIÓN POR PARTICIÓN**

| partition  | n_total | n_bad | n_good | auc      | gini     | ks       | ks_cutoff_risk_score | ks_cutoff_score | tpr_at_ks | fpr_at_ks | source        | status |
|------------|---------|-------|--------|----------|----------|----------|----------------------|-----------------|-----------|-----------|---------------|--------|
| desarrollo | 3961    | 924   | 3037   | 0.712346 | 0.424692 | 0.320144 | 0.189651             | —               | 0.706710  | 0.386566  | pd_calibrated | ok     |
| holdout    | 1031    | 244   | 787    | 0.694646 | 0.389292 | 0.311892 | 0.223207             | —               | 0.590164  | 0.278272  | pd_calibrated | ok     |
| oot        | 1008    | 239   | 769    | 0.656096 | 0.312192 | 0.251906 | 0.259595             | —               | 0.456067  | 0.204161  | pd_calibrated | ok     |

## DESEMPEÑO POR DECIL DE SCORE

| partition  | decile | n_total | n_bad | n_good | bad_rate | good_rate | mean_pd  | min_pd   | max_pd   | mean_score | min_score  | max_score  | cum_total | cum_bad | cum_good | cum_bad_capture_rate | cum_good_capture_rate | lift     | ks_at_decile |
|------------|--------|---------|-------|--------|----------|-----------|----------|----------|----------|------------|------------|------------|-----------|---------|----------|----------------------|-----------------------|----------|--------------|
| desarrollo | 1      | 397     | 203   | 194    | 0.5113   | 0.4887    | 0.464277 | 0.368829 | 0.774565 | 484.874055 | 446.000000 | 497.000000 | 397       | 203     | 194      | 0.2197               | 0.0639                | 2.191989 | 0.155818     |
| desarrollo | 2      | 396     | 154   | 242    | 0.3889   | 0.6111    | 0.326896 | 0.293402 | 0.368829 | 501.558081 | 496.000000 | 507.000000 | 793       | 357     | 436      | 0.3864               | 0.1436                | 1.667088 | 0.242801     |
| desarrollo | 3      | 396     | 129   | 267    | 0.3258   | 0.6742    | 0.269114 | 0.245896 | 0.293402 | 509.434343 | 505.000000 | 514.000000 | 1189      | 486     | 703      | 0.5260               | 0.2315                | 1.396456 | 0.294496     |
| desarrollo | 4      | 396     | 99    | 297    | 0.2500   | 0.7500    | 0.227482 | 0.208969 | 0.245896 | 515.858586 | 512.000000 | 520.000000 | 1585      | 585     | 1000     | 0.6331               | 0.3293                | 1.071699 | 0.303845     |
| desarrollo | 5      | 396     | 93    | 303    | 0.2348   | 0.7652    | 0.192104 | 0.176307 | 0.208969 | 521.957071 | 518.000000 | 526.000000 | 1981      | 678     | 1303     | 0.7338               | 0.4290                | 1.006748 | 0.304724     |
| desarrollo | 6      | 396     | 67    | 329    | 0.1692   | 0.8308    | 0.160814 | 0.147936 | 0.176307 | 528.070707 | 524.000000 | 532.000000 | 2377      | 745     | 1632     | 0.8063               | 0.5374                | 0.725291 | 0.268905     |
| desarrollo | 7      | 396     | 65    | 331    | 0.1641   | 0.8359    | 0.133788 | 0.120596 | 0.147936 | 534.315657 | 530.000000 | 539.000000 | 2773      | 810     | 1963     | 0.8766               | 0.6464                | 0.703641 | 0.230262     |
| desarrollo | 8      | 396     | 54    | 342    | 0.1364   | 0.8636    | 0.105434 | 0.090212 | 0.120596 | 542.138889 | 538.000000 | 548.000000 | 3169      | 864     | 2305     | 0.9351               | 0.7590                | 0.584563 | 0.176092     |
| desarrollo | 9      | 396     | 44    | 352    | 0.1111   | 0.8889    | 0.076569 | 0.061318 | 0.090107 | 552.510101 | 546.000000 | 560.000000 | 3565      | 908     | 2657     | 0.9827               | 0.8749                | 0.476311 | 0.107807     |
| desarrollo | 10     | 396     | 16    | 380    | 0.0404   | 0.9596    | 0.042856 | 0.008485 | 0.061318 | 571.585859 | 558.000000 | 618.000000 | 3961      | 924     | 3037     | 1.0000               | 1.0000                | 0.173204 | 0.000000     |
| holdout    | 1      | 104     | 52    | 52     | 0.5000   | 0.5000    | 0.480290 | 0.368464 | 0.732205 | 482.942308 | 452.000000 | 496.000000 | 104       | 52      | 52       | 0.2131               | 0.0661                | 2.112705 | 0.147041     |
| holdout    | 2      | 103     | 41    | 62     | 0.3981   | 0.6019    | 0.332514 | 0.298887 | 0.368464 | 500.737864 | 496.000000 | 506.000000 | 207       | 93      | 114      | 0.3811               | 0.1449                | 1.681959 | 0.236294     |
| holdout    | 3      | 103     | 34    | 69     | 0.3301   | 0.6699    | 0.274019 | 0.250344 | 0.298887 | 508.689320 | 504.000000 | 513.000000 | 310       | 127     | 183      | 0.5205               | 0.2325                | 1.394795 | 0.287963     |
| holdout    | 4      | 103     | 25    | 78     | 0.2427   | 0.7573    | 0.225597 | 0.204709 | 0.250344 | 516.155340 | 511.000000 | 520.000000 | 413       | 152     | 261      | 0.6230               | 0.3316                | 1.025585 | 0.291312     |
| holdout    | 5      | 103     | 22    | 81     | 0.2136   | 0.7864    | 0.188267 | 0.173891 | 0.204661 | 522.592233 | 519.000000 | 526.000000 | 516       | 174     | 342      | 0.7131               | 0.4346                | 0.902515 | 0.278553     |
| holdout    | 6      | 103     | 13    | 90     | 0.1262   | 0.8738    | 0.162510 | 0.152213 | 0.173846 | 527.669903 | 525.000000 | 531.000000 | 619       | 187     | 432      | 0.7664               | 0.5489                | 0.533304 | 0.217473     |
| holdout    | 7      | 103     | 23    | 80     | 0.2233   | 0.7767    | 0.138393 | 0.124609 | 0.151995 | 533.223301 | 530.000000 | 538.000000 | 722       | 210     | 512      | 0.8607               | 0.6506                | 0.943538 | 0.210084     |
| holdout    | 8      | 103     | 15    | 88     | 0.1456   | 0.8544    | 0.106735 | 0.090472 | 0.124343 | 541.864078 | 536.000000 | 547.000000 | 825       | 225     | 600      | 0.9221               | 0.7624                | 0.615351 | 0.159742     |
| holdout    | 9      | 103     | 10    | 93     | 0.0971   | 0.9029    | 0.076171 | 0.059697 | 0.090212 | 552.660194 | 546.000000 | 560.000000 | 928       | 235     | 693      | 0.9631               | 0.8806                | 0.410234 | 0.082556     |
| holdout    | 10     | 103     | 9     | 94     | 0.0874   | 0.9126    | 0.044013 | 0.014361 | 0.059655 | 570.446602 | 560.000000 | 603.000000 | 1031      | 244     | 787      | 1.0000               | 1.0000                | 0.369211 | 0.000000     |
| oot        | 1      | 101     | 45    | 56     | 0.4455   | 0.5545    | 0.480221 | 0.376923 | 0.732205 | 482.920792 | 452.000000 | 496.000000 | 101       | 45      | 56       | 0.1883               | 0.0728                | 1.879117 | 0.115463     |
| oot        | 2      | 101     | 38    | 63     | 0.3762   | 0.6238    | 0.329443 | 0.291975 | 0.376607 | 501.188119 | 494.000000 | 507.000000 | 202       | 83      | 119      | 0.3473               | 0.1547                | 1.586810 | 0.192534     |
| oot        | 3      | 101     | 32    | 69     | 0.3168   | 0.6832    | 0.265686 | 0.242490 | 0.291014 | 509.930693 | 506.000000 | 514.000000 | 303       | 115     | 188      | 0.4812               | 0.2445                | 1.336261 | 0.236698     |
| oot        | 4      | 101     | 17    | 84     | 0.1683   | 0.8317    | 0.224310 | 0.203208 | 0.242120 | 516.405941 | 513.000000 | 520.000000 | 404       | 132     | 272      | 0.5523               | 0.3537                | 0.709889 | 0.198595     |
| oot        | 5      | 101     | 26    | 75     | 0.2574   | 0.7426    | 0.184714 | 0.168869 | 0.202989 | 523.277228 | 519.000000 | 527.000000 | 505       | 158     | 347      | 0.6611               | 0.4512                | 1.085712 | 0.209852     |
| oot        | 6      | 101     | 23    | 78     | 0.2277   | 0.7723    | 0.157103 | 0.144751 | 0.168869 | 529.069307 | 526.000000 | 533.000000 | 606       | 181     | 425      | 0.7573               | 0.5527                | 0.960437 | 0.204656     |
| oot        | 7      | 101     | 15    | 86     | 0.1485   | 0.8515    | 0.132068 | 0.120596 | 0.144751 | 534.732673 | 532.000000 | 538.000000 | 707       | 196     | 511      | 0.8201               | 0.6645                | 0.626372 | 0.155584     |
| oot        | 8      | 101     | 20    | 81     | 0.1980   | 0.8020    | 0.106984 | 0.092326 | 0.119822 | 541.574257 | 538.000000 | 547.000000 | 808       | 216     | 592      | 0.9038               | 0.7698                | 0.835163 | 0.133935     |
| oot        | 9      | 100     | 13    | 87     | 0.1300   | 0.8700    | 0.076182 | 0.061411 | 0.091239 | 552.610000 | 546.000000 | 559.000000 | 908       | 229     | 679      | 0.9582               | 0.8830                | 0.548285 | 0.075194     |
| oot        | 10     | 100     | 10    | 90     | 0.1000   | 0.9000    | 0.043492 | 0.007587 | 0.061318 | 571.250000 | 558.000000 | 622.000000 | 1008      | 239     | 769      | 1.0000               | 1.0000                | 0.421757 | 0.000000     |

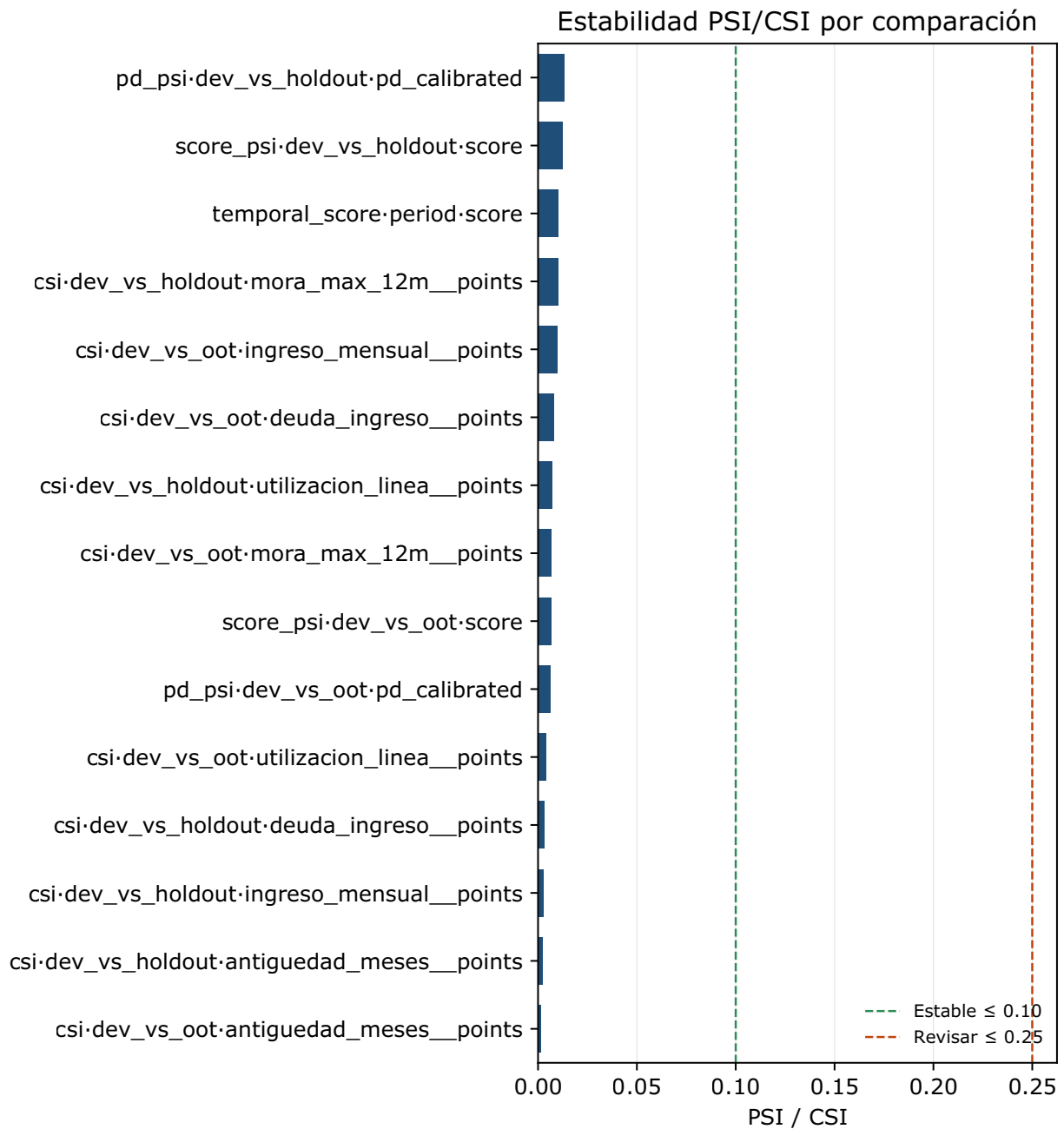
## 4.7 Estabilidad

El PSI del score en Desarrollo vs. Holdout es 0,0132 (banda «Estable»). La distribución del score se mantiene, de modo que no hay evidencia de desplazamiento poblacional.

El PSI del score en Desarrollo vs. OOT es 0,0068 (banda «Estable»). La distribución del score se mantiene, de modo que no hay evidencia de desplazamiento poblacional.

A nivel de variable, el mayor CSI corresponde a «mora\_max\_12m\_\_points» con 0,0102.

El PSI se calculó sobre 10 tramos del score, definidos en Desarrollo.



### MÉTRICAS DE ESTABILIDAD (PSI/CSI)

| metric    | comparison     | feature                  | value    | stable_threshold | review_threshold | band   | action |
|-----------|----------------|--------------------------|----------|------------------|------------------|--------|--------|
| score_psi | dev_vs_holdout | score                    | 0.012661 | 0.100000         | 0.250000         | stable | —      |
| score_psi | dev_vs_oot     | score                    | 0.006792 | 0.100000         | 0.250000         | stable | —      |
| pd_psi    | dev_vs_holdout | pd_calibrated            | 0.013182 | 0.100000         | 0.250000         | stable | —      |
| pd_psi    | dev_vs_oot     | pd_calibrated            | 0.006465 | 0.100000         | 0.250000         | stable | —      |
| csi       | dev_vs_holdout | antiguedad_meses__points | 0.002587 | 0.100000         | 0.250000         | stable | —      |
| csi       | dev_vs_oot     | antiguedad_meses__points | 0.001531 | 0.100000         | 0.250000         | stable | —      |



| metric         | comparison     | feature                       | value    | stable_threshold | review_threshold | band   | action |
|----------------|----------------|-------------------------------|----------|------------------|------------------|--------|--------|
| csi            | dev_vs_holdout | deuda_ingreso_<br>_points     | 0.003322 | 0.100000         | 0.250000         | stable | —      |
| csi            | dev_vs_oot     | deuda_ingreso_<br>_points     | 0.008178 | 0.100000         | 0.250000         | stable | —      |
| csi            | dev_vs_holdout | ingreso_mensu<br>al__points   | 0.002994 | 0.100000         | 0.250000         | stable | —      |
| csi            | dev_vs_oot     | ingreso_mensu<br>al__points   | 0.009808 | 0.100000         | 0.250000         | stable | —      |
| csi            | dev_vs_holdout | mora_max_12m<br>__points      | 0.010209 | 0.100000         | 0.250000         | stable | —      |
| csi            | dev_vs_oot     | mora_max_12m<br>__points      | 0.006914 | 0.100000         | 0.250000         | stable | —      |
| csi            | dev_vs_holdout | utilizacion_linea<br>__points | 0.007215 | 0.100000         | 0.250000         | stable | —      |
| csi            | dev_vs_oot     | utilizacion_linea<br>__points | 0.003973 | 0.100000         | 0.250000         | stable | —      |
| temporal_score | period         | score                         | 0.010497 | 0.100000         | 0.250000         | stable | —      |

## PSI POR TRAMO DE SCORE

| metric    | comparison     | feature       | bin_label | expected_count | actual_count | expected_pct | actual_pct | component_value | total_value | band   |
|-----------|----------------|---------------|-----------|----------------|--------------|--------------|------------|-----------------|-------------|--------|
| score_psi | dev_vs_holdout | score         | bin_00    | 376            | 99           | 0.0949       | 0.0960     | 0.000013        | 0.012661    | stable |
| score_psi | dev_vs_holdout | score         | bin_01    | 393            | 113          | 0.0992       | 0.1096     | 0.001034        | 0.012661    | stable |
| score_psi | dev_vs_holdout | score         | bin_02    | 388            | 102          | 0.0980       | 0.0989     | 0.000010        | 0.012661    | stable |
| score_psi | dev_vs_holdout | score         | bin_03    | 412            | 85           | 0.1040       | 0.0824     | 0.005013        | 0.012661    | stable |
| score_psi | dev_vs_holdout | score         | bin_04    | 368            | 97           | 0.0929       | 0.0941     | 0.000015        | 0.012661    | stable |
| score_psi | dev_vs_holdout | score         | bin_05    | 409            | 129          | 0.1033       | 0.1251     | 0.004199        | 0.012661    | stable |
| score_psi | dev_vs_holdout | score         | bin_06    | 400            | 105          | 0.1010       | 0.1018     | 0.000007        | 0.012661    | stable |
| score_psi | dev_vs_holdout | score         | bin_07    | 405            | 90           | 0.1022       | 0.0873     | 0.002364        | 0.012661    | stable |
| score_psi | dev_vs_holdout | score         | bin_08    | 390            | 101          | 0.0985       | 0.0980     | 0.000003        | 0.012661    | stable |
| score_psi | dev_vs_holdout | score         | bin_09    | 420            | 110          | 0.1060       | 0.1067     | 0.000004        | 0.012661    | stable |
| score_psi | dev_vs_oot     | score         | bin_00    | 376            | 102          | 0.0949       | 0.1012     | 0.000400        | 0.006792    | stable |
| score_psi | dev_vs_oot     | score         | bin_01    | 393            | 93           | 0.0992       | 0.0923     | 0.000506        | 0.006792    | stable |
| score_psi | dev_vs_oot     | score         | bin_02    | 388            | 96           | 0.0980       | 0.0952     | 0.000076        | 0.006792    | stable |
| score_psi | dev_vs_oot     | score         | bin_03    | 412            | 94           | 0.1040       | 0.0933     | 0.001175        | 0.006792    | stable |
| score_psi | dev_vs_oot     | score         | bin_04    | 368            | 86           | 0.0929       | 0.0853     | 0.000647        | 0.006792    | stable |
| score_psi | dev_vs_oot     | score         | bin_05    | 409            | 112          | 0.1033       | 0.1111     | 0.000576        | 0.006792    | stable |
| score_psi | dev_vs_oot     | score         | bin_06    | 400            | 114          | 0.1010       | 0.1131     | 0.001372        | 0.006792    | stable |
| score_psi | dev_vs_oot     | score         | bin_07    | 405            | 115          | 0.1022       | 0.1141     | 0.001297        | 0.006792    | stable |
| score_psi | dev_vs_oot     | score         | bin_08    | 390            | 91           | 0.0985       | 0.0903     | 0.000710        | 0.006792    | stable |
| score_psi | dev_vs_oot     | score         | bin_09    | 420            | 105          | 0.1060       | 0.1042     | 0.000033        | 0.006792    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_00    | 395            | 107          | 0.0997       | 0.1038     | 0.000162        | 0.013182    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_01    | 397            | 98           | 0.1002       | 0.0951     | 0.000274        | 0.013182    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_02    | 393            | 91           | 0.0992       | 0.0883     | 0.001281        | 0.013182    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_03    | 397            | 99           | 0.1002       | 0.0960     | 0.000180        | 0.013182    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_04    | 397            | 129          | 0.1002       | 0.1251     | 0.005523        | 0.013182    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_05    | 390            | 104          | 0.0985       | 0.1009     | 0.000058        | 0.013182    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_06    | 401            | 83           | 0.1012       | 0.0805     | 0.004751        | 0.013182    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_07    | 396            | 105          | 0.1000       | 0.1018     | 0.000035        | 0.013182    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_08    | 396            | 113          | 0.1000       | 0.1096     | 0.000885        | 0.013182    | stable |
| pd_psi    | dev_vs_holdout | pd_calibrated | bin_09    | 399            | 102          | 0.1007       | 0.0989     | 0.000032        | 0.013182    | stable |
| pd_psi    | dev_vs_oot     | pd_calibrated | bin_00    | 395            | 99           | 0.0997       | 0.0982     | 0.000023        | 0.006465    | stable |

| metric | comparison     | feature                  | bin_label | expected_count | actual_count | expected_pct | actual_pct | component_value | total_value | band   |
|--------|----------------|--------------------------|-----------|----------------|--------------|--------------|------------|-----------------|-------------|--------|
| pd_psi | dev_vs_oot     | pd_calibrated            | bin_01    | 397            | 96           | 0.1002       | 0.0952     | 0.000255        | 0.006465    | stable |
| pd_psi | dev_vs_oot     | pd_calibrated            | bin_02    | 393            | 106          | 0.0992       | 0.1052     | 0.000346        | 0.006465    | stable |
| pd_psi | dev_vs_oot     | pd_calibrated            | bin_03    | 397            | 112          | 0.1002       | 0.1111     | 0.001122        | 0.006465    | stable |
| pd_psi | dev_vs_oot     | pd_calibrated            | bin_04    | 397            | 113          | 0.1002       | 0.1121     | 0.001330        | 0.006465    | stable |
| pd_psi | dev_vs_oot     | pd_calibrated            | bin_05    | 390            | 89           | 0.0985       | 0.0883     | 0.001108        | 0.006465    | stable |
| pd_psi | dev_vs_oot     | pd_calibrated            | bin_06    | 401            | 98           | 0.1012       | 0.0972     | 0.000162        | 0.006465    | stable |
| pd_psi | dev_vs_oot     | pd_calibrated            | bin_07    | 396            | 97           | 0.1000       | 0.0962     | 0.000143        | 0.006465    | stable |
| pd_psi | dev_vs_oot     | pd_calibrated            | bin_08    | 396            | 89           | 0.1000       | 0.0883     | 0.001451        | 0.006465    | stable |
| pd_psi | dev_vs_oot     | pd_calibrated            | bin_09    | 399            | 109          | 0.1007       | 0.1081     | 0.000525        | 0.006465    | stable |
| csi    | dev_vs_holdout | antiguedad_meses__points | pts=94.0  | 384            | 93           | 0.0969       | 0.0902     | 0.000486        | 0.002587    | stable |
| csi    | dev_vs_holdout | antiguedad_meses__points | pts=100.0 | 763            | 211          | 0.1926       | 0.2047     | 0.000728        | 0.002587    | stable |
| csi    | dev_vs_holdout | antiguedad_meses__points | pts=104.0 | 1769           | 453          | 0.4466       | 0.4394     | 0.000118        | 0.002587    | stable |
| csi    | dev_vs_holdout | antiguedad_meses__points | pts=109.0 | 838            | 227          | 0.2116       | 0.2202     | 0.000344        | 0.002587    | stable |
| csi    | dev_vs_holdout | antiguedad_meses__points | pts=128.0 | 207            | 47           | 0.0523       | 0.0456     | 0.000912        | 0.002587    | stable |
| csi    | dev_vs_oot     | antiguedad_meses__points | pts=94.0  | 384            | 90           | 0.0969       | 0.0893     | 0.000630        | 0.001531    | stable |
| csi    | dev_vs_oot     | antiguedad_meses__points | pts=100.0 | 763            | 203          | 0.1926       | 0.2014     | 0.000390        | 0.001531    | stable |
| csi    | dev_vs_oot     | antiguedad_meses__points | pts=104.0 | 1769           | 446          | 0.4466       | 0.4425     | 0.000039        | 0.001531    | stable |
| csi    | dev_vs_oot     | antiguedad_meses__points | pts=109.0 | 838            | 220          | 0.2116       | 0.2183     | 0.000208        | 0.001531    | stable |
| csi    | dev_vs_oot     | antiguedad_meses__points | pts=128.0 | 207            | 49           | 0.0523       | 0.0486     | 0.000264        | 0.001531    | stable |
| csi    | dev_vs_holdout | deuda_ingreso__points    | pts=71.0  | 205            | 52           | 0.0518       | 0.0504     | 0.000034        | 0.003322    | stable |
| csi    | dev_vs_holdout | deuda_ingreso__points    | pts=92.0  | 674            | 195          | 0.1702       | 0.1891     | 0.002007        | 0.003322    | stable |
| csi    | dev_vs_holdout | deuda_ingreso__points    | pts=101.0 | 654            | 156          | 0.1651       | 0.1513     | 0.001205        | 0.003322    | stable |
| csi    | dev_vs_holdout | deuda_ingreso__points    | pts=111.0 | 1577           | 410          | 0.3981       | 0.3977     | 0.000001        | 0.003322    | stable |
| csi    | dev_vs_holdout | deuda_ingreso__points    | pts=113.0 | 558            | 144          | 0.1409       | 0.1397     | 0.000010        | 0.003322    | stable |
| csi    | dev_vs_holdout | deuda_ingreso__points    | pts=127.0 | 293            | 74           | 0.0740       | 0.0718     | 0.000066        | 0.003322    | stable |
| csi    | dev_vs_oot     | deuda_ingreso__points    | pts=71.0  | 205            | 58           | 0.0518       | 0.0575     | 0.000613        | 0.008178    | stable |
| csi    | dev_vs_oot     | deuda_ingreso__points    | pts=92.0  | 674            | 186          | 0.1702       | 0.1845     | 0.001164        | 0.008178    | stable |
| csi    | dev_vs_oot     | deuda_ingreso__points    | pts=101.0 | 654            | 149          | 0.1651       | 0.1478     | 0.001913        | 0.008178    | stable |
| csi    | dev_vs_oot     | deuda_ingreso__points    | pts=111.0 | 1577           | 413          | 0.3981       | 0.4097     | 0.000333        | 0.008178    | stable |
| csi    | dev_vs_oot     | deuda_ingreso__points    | pts=113.0 | 558            | 144          | 0.1409       | 0.1429     | 0.000028        | 0.008178    | stable |
| csi    | dev_vs_oot     | deuda_ingreso__points    | pts=127.0 | 293            | 58           | 0.0740       | 0.0575     | 0.004128        | 0.008178    | stable |
| csi    | dev_vs_holdout | ingreso_mensual__points  | pts=77.0  | 210            | 63           | 0.0530       | 0.0611     | 0.001149        | 0.002994    | stable |
| csi    | dev_vs_holdout | ingreso_mensual__points  | pts=87.0  | 587            | 154          | 0.1482       | 0.1494     | 0.000009        | 0.002994    | stable |
| csi    | dev_vs_holdout | ingreso_mensual__points  | pts=97.0  | 806            | 202          | 0.2035       | 0.1959     | 0.000286        | 0.002994    | stable |

| metric | comparison     | feature                   | bin_label | expected_count | actual_count | expected_pct | actual_pct | component_value | total_value | band   |
|--------|----------------|---------------------------|-----------|----------------|--------------|--------------|------------|-----------------|-------------|--------|
| csi    | dev_vs_holdout | ingreso_mensual__points   | pts=105.0 | 917            | 223          | 0.2315       | 0.2163     | 0.001034        | 0.002994    | stable |
| csi    | dev_vs_holdout | ingreso_mensual__points   | pts=117.0 | 876            | 238          | 0.2212       | 0.2308     | 0.000415        | 0.002994    | stable |
| csi    | dev_vs_holdout | ingreso_mensual__points   | pts=141.0 | 565            | 151          | 0.1426       | 0.1465     | 0.000101        | 0.002994    | stable |
| csi    | dev_vs_oot     | ingreso_mensual__points   | pts=77.0  | 210            | 61           | 0.0530       | 0.0605     | 0.000992        | 0.009808    | stable |
| csi    | dev_vs_oot     | ingreso_mensual__points   | pts=87.0  | 587            | 144          | 0.1482       | 0.1429     | 0.000196        | 0.009808    | stable |
| csi    | dev_vs_oot     | ingreso_mensual__points   | pts=97.0  | 806            | 170          | 0.2035       | 0.1687     | 0.006540        | 0.009808    | stable |
| csi    | dev_vs_oot     | ingreso_mensual__points   | pts=105.0 | 917            | 239          | 0.2315       | 0.2371     | 0.000134        | 0.009808    | stable |
| csi    | dev_vs_oot     | ingreso_mensual__points   | pts=117.0 | 876            | 239          | 0.2212       | 0.2371     | 0.001110        | 0.009808    | stable |
| csi    | dev_vs_oot     | ingreso_mensual__points   | pts=141.0 | 565            | 155          | 0.1426       | 0.1538     | 0.000836        | 0.009808    | stable |
| csi    | dev_vs_holdout | mora_max_12m__points      | pts=99.0  | 349            | 94           | 0.0881       | 0.0912     | 0.000105        | 0.010209    | stable |
| csi    | dev_vs_holdout | mora_max_12m__points      | pts=101.0 | 678            | 162          | 0.1712       | 0.1571     | 0.001202        | 0.010209    | stable |
| csi    | dev_vs_holdout | mora_max_12m__points      | pts=102.0 | 1166           | 340          | 0.2944       | 0.3298     | 0.004021        | 0.010209    | stable |
| csi    | dev_vs_holdout | mora_max_12m__points      | pts=107.0 | 1173           | 273          | 0.2961       | 0.2648     | 0.003507        | 0.010209    | stable |
| csi    | dev_vs_holdout | mora_max_12m__points      | pts=110.0 | 355            | 90           | 0.0896       | 0.0873     | 0.000061        | 0.010209    | stable |
| csi    | dev_vs_holdout | mora_max_12m__points      | pts=114.0 | 240            | 72           | 0.0606       | 0.0698     | 0.001313        | 0.010209    | stable |
| csi    | dev_vs_oot     | mora_max_12m__points      | pts=99.0  | 349            | 97           | 0.0881       | 0.0962     | 0.000716        | 0.006914    | stable |
| csi    | dev_vs_oot     | mora_max_12m__points      | pts=101.0 | 678            | 174          | 0.1712       | 0.1726     | 0.000012        | 0.006914    | stable |
| csi    | dev_vs_oot     | mora_max_12m__points      | pts=102.0 | 1166           | 296          | 0.2944       | 0.2937     | 0.000002        | 0.006914    | stable |
| csi    | dev_vs_oot     | mora_max_12m__points      | pts=107.0 | 1173           | 307          | 0.2961       | 0.3046     | 0.000236        | 0.006914    | stable |
| csi    | dev_vs_oot     | mora_max_12m__points      | pts=110.0 | 355            | 69           | 0.0896       | 0.0685     | 0.005705        | 0.006914    | stable |
| csi    | dev_vs_oot     | mora_max_12m__points      | pts=114.0 | 240            | 65           | 0.0606       | 0.0645     | 0.000242        | 0.006914    | stable |
| csi    | dev_vs_holdout | utilizacion_linea__points | pts=89.0  | 343            | 113          | 0.0866       | 0.1096     | 0.005421        | 0.007215    | stable |
| csi    | dev_vs_holdout | utilizacion_linea__points | pts=99.0  | 1212           | 308          | 0.3060       | 0.2987     | 0.000174        | 0.007215    | stable |
| csi    | dev_vs_holdout | utilizacion_linea__points | pts=106.0 | 1385           | 338          | 0.3497       | 0.3278     | 0.001406        | 0.007215    | stable |
| csi    | dev_vs_holdout | utilizacion_linea__points | pts=112.0 | 788            | 208          | 0.1989       | 0.2017     | 0.000039        | 0.007215    | stable |
| csi    | dev_vs_holdout | utilizacion_linea__points | pts=124.0 | 233            | 64           | 0.0588       | 0.0621     | 0.000175        | 0.007215    | stable |
| csi    | dev_vs_oot     | utilizacion_linea__points | pts=89.0  | 343            | 101          | 0.0866       | 0.1002     | 0.001985        | 0.003973    | stable |
| csi    | dev_vs_oot     | utilizacion_linea__points | pts=99.0  | 1212           | 298          | 0.3060       | 0.2956     | 0.000356        | 0.003973    | stable |
| csi    | dev_vs_oot     | utilizacion_linea__points | pts=106.0 | 1385           | 335          | 0.3497       | 0.3323     | 0.000880        | 0.003973    | stable |
| csi    | dev_vs_oot     | utilizacion_linea__points | pts=112.0 | 788            | 212          | 0.1989       | 0.2103     | 0.000633        | 0.003973    | stable |
| csi    | dev_vs_oot     | utilizacion_linea__points | pts=124.0 | 233            | 62           | 0.0588       | 0.0615     | 0.000120        | 0.003973    | stable |

## 5 Validación formal

La validación formal del modelo «preset-estandar-consumo» ejecutó discriminación, calibración y estabilidad. El estado técnico agregado publicado por el motor es «fail».

El resultado registra 3 tests, de los cuales 3 quedaron fallidos. Las tablas se copian del ValidationResult atómico, sin recalcular métricas ni decisiones.

Este estado es evidencia técnica; el veredicto de aprobación, aprobación con observaciones o rechazo corresponde al validador humano y queda POR COMPLETAR.

### POR COMPLETAR — VEREDICTO DE VALIDACIÓN FORMAL

Emite el juicio humano sobre la aptitud del modelo para su uso previsto: aprobar, aprobar con observaciones o rechazar.

Documenta observaciones, responsables, plazos y la fecha de la próxima revisión; el estado técnico del motor no sustituye este veredicto.

### 5.1 Discriminación

La tabla reproduce por partición AUC, Gini y KS, junto con población, fuente y estado de evaluabilidad. Filas publicadas por el DTO: 3.

#### VALIDACIÓN FORMAL — DISCRIMINACIÓN

| partition  | n_total | n_bad | auc      | gini     | ks       | source               | status |
|------------|---------|-------|----------|----------|----------|----------------------|--------|
| desarrollo | 3961    | 924   | 0.712346 | 0.424692 | 0.320144 | performance_artifact | ok     |
| holdout    | 1031    | 244   | 0.694646 | 0.389292 | 0.311892 | performance_artifact | ok     |
| oot        | 1008    | 239   | 0.656096 | 0.312192 | 0.251906 | performance_artifact | ok     |

### 5.2 Calibración

La tabla reúne Hosmer-Lemeshow, Brier y, cuando fue configurado, el contraste por grado con sus decisiones y semáforo publicados. Filas publicadas por el DTO: 6.

## VALIDACIÓN FORMAL — CALIBRACIÓN

| partition  | test            | grade | n    | observed_defaults | expected_pd | observed_dr | statistic | degrees_of_freedom | p_value  | alpha    | decision      | traffic_light |
|------------|-----------------|-------|------|-------------------|-------------|-------------|-----------|--------------------|----------|----------|---------------|---------------|
| desarrollo | hosmer_lemeshow | ALL   | 3961 | 924               | 0.200000    | 0.233274    | 37.243376 | 8                  | 0.000010 | 0.050000 | fail          | —             |
| desarrollo | brier           | ALL   | 3961 | 924               | 0.200000    | 0.233274    | 0.161492  | —                  | —        | —        | not_evaluable | —             |
| oot        | hosmer_lemeshow | ALL   | 1008 | 239               | 0.200298    | 0.237103    | 32.223459 | 8                  | 0.000085 | 0.050000 | fail          | —             |
| oot        | brier           | ALL   | 1008 | 239               | 0.200298    | 0.237103    | 0.171861  | —                  | —        | —        | not_evaluable | —             |
| holdout    | hosmer_lemeshow | ALL   | 1031 | 244               | 0.203120    | 0.236663    | 19.983637 | 8                  | 0.010398 | 0.050000 | fail          | —             |
| holdout    | brier           | ALL   | 1031 | 244               | 0.203120    | 0.236663    | 0.165269  | —                  | —        | —        | not_evaluable | —             |

## 5.3 Estabilidad

La tabla reproduce PSI/estabilidad, umbrales, banda, acción y decisión publicados por validation. Filas publicadas por el DTO: 15.

## VALIDACIÓN FORMAL — ESTABILIDAD

| metric         | comparison     | feature                   | value    | stable_threshold | review_threshold | band   | action | source             | status | decision |
|----------------|----------------|---------------------------|----------|------------------|------------------|--------|--------|--------------------|--------|----------|
| score_psi      | dev_vs_holdout | score                     | 0.012661 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| score_psi      | dev_vs_oout    | score                     | 0.006792 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| pd_psi         | dev_vs_holdout | pd_calibrated             | 0.013182 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| pd_psi         | dev_vs_oout    | pd_calibrated             | 0.006465 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_holdout | antiguedad_meses__points  | 0.002587 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_oout    | antiguedad_meses__points  | 0.001531 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_holdout | deuda_ingreso__points     | 0.003322 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_oout    | deuda_ingreso__points     | 0.008178 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_holdout | ingreso_mensual__points   | 0.002994 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_oout    | ingreso_mensual__points   | 0.009808 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_holdout | mora_max_12m__points      | 0.010209 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_oout    | mora_max_12m__points      | 0.006914 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_holdout | utilizacion_linea__points | 0.007215 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| csi            | dev_vs_oout    | utilizacion_linea__points | 0.003973 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |
| temporal_score | period         | score                     | 0.010497 | 0.100000         | 0.250000         | stable | —      | stability_artifact | ok     | pass     |

## 6 Conclusiones y recomendación

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Discriminación (AUC): Desarrollo 0,7123, Holdout 0,6946 y Fuera de tiempo (OOT) 0,6561.

Estabilidad (PSI del score): Desarrollo vs. Holdout 0,0132 («Estable») y Desarrollo vs. OOT 0,0068 («Estable»).

Calibración: la PD media calibrada en Desarrollo (20,00 %) se contrasta con la tasa observada (23,33 %).

Estos hallazgos son los que el motor puede sostener con los números de la corrida. La recomendación —aprobar, aprobar con observaciones o rechazar— es un juicio humano y no se deriva automáticamente de ellos.

### **POR COMPLETAR — CONCLUSIONES Y RECOMENDACIÓN**

Emite el juicio: ¿el modelo es apto para el uso previsto? La recomendación (aprobar, aprobar con observaciones o rechazar) la firma un humano, no el motor.

Enumera las observaciones y condiciones de uso, con responsable y plazo para cada una, y fija la fecha de la próxima revisión.



## 7 Limitaciones y supuestos

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El alcance de este informe es la construcción y evaluación del scorecard. La validación formal ejecutó discriminación, calibración y estabilidad y se documenta en su capítulo propio; el estado técnico no sustituye el veredicto humano.

Todas las métricas provienen de la corrida trazada en el Anexo A. El informe no completa ningún hueco con supuestos: lo que no se pudo calcular aparece declarado como no disponible.

Caveats de reproducibilidad declarados por la corrida: working tree git sucio: cambios sin commitear no reconstruibles.

No hay secciones obligatorias ausentes en esta corrida.

## Anexo A — Lineage y reproducibilidad

---

La corrida queda identificada por los hashes de configuración y de datos, el commit del código y la semilla raíz. Con estos cuatro valores el resultado es reproducible.

### PARÁMETROS Y PAYLOAD

|                            |                                                                                                                                                   |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>config_hash</b>         | 4b93809debd102cf1644f8b5acaa9abbd693204e9f3fb7346f392425677ae49f                                                                                  |
| <b>created_at</b>          | 2026-07-21T01:52:21.484318Z                                                                                                                       |
| <b>data_hash</b>           | 74c5d9ddeca8ff71853623d20ace015d0788564549339d61bfb66f24dfadfcc8                                                                                  |
| <b>determinism_caveats</b> | <pre>[<br/>  "working tree git sucio: cambios sin commitear no<br/>  reconstruibles"<br/>]</pre>                                                  |
| <b>git_dirty</b>           | true                                                                                                                                              |
| <b>git_sha</b>             | ad5fe58dc5eededee97a4d4d7c3c4b8ef979af                                                                                                            |
| <b>library_versions</b>    | <pre>{<br/>  "PyYAML": "6.0.3",<br/>  "nikodym": "1.4.0",<br/>  "numpy": "2.4.6",<br/>  "pandas": "2.3.3",<br/>  "pydantic": "2.13.4"<br/>}</pre> |
| <b>root_seed</b>           | 20240706                                                                                                                                          |
| <b>schema_version</b>      | 1.0.0                                                                                                                                             |
| <b>uv_lock_hash</b>        | —                                                                                                                                                 |

## Anexo B — Tablas detalladas

Este anexo reproduce íntegras todas las tablas que produjo la corrida, incluidas las que el cuerpo del informe ya mostró. Es la trazabilidad completa: nada de lo que el motor calculó queda fuera del documento.

### PARÁMETROS Y PAYLOAD

**figure\_keys**

[]

**table\_keys**

```
[
  "binning.tables.antiguedad_meses",
  "binning.tables.deuda_ingreso",
  "binning.tables.ingreso_mensual",
  "binning.tables.mora_max_12m",
  "binning.tables.segmento",
  "binning.tables.utilizacion_linea",
  "binning.summary",
  "binning.woe_frame",
  "selection.selected_woe_frame",
  "selection.selection_table",
  "selection.correlation_matrix",
  "selection.vif_table",
  "selection.stability_table",
  "model.coefficients",
  "model.raw_pd_frame",
  "scorecard.scorecard",
  "scorecard.score",
  "calibration.calibrated_pd_frame",
  "performance.performance_table",
  "performance.discriminant_metrics",
  "stability.psi_table",
  "stability.stability_metrics",
  "data.states",
  "data.partitions",
  "data.exclusions",
  "validation.discrimination",
  "validation.calibration",
  "validation.stability"
]
```

### TABLA DE BINNING WOE — VARIABLE «ANTIGUEDAD\_MESES»

| Bin             | Count | Count (%) | Non-event | Event | Event rate | WoE       | IV       | JS       |
|-----------------|-------|-----------|-----------|-------|------------|-----------|----------|----------|
| (-inf, 12.50)   | 384   | 0.096945  | 271       | 113   | 0.2943     | -0.315182 | 0.010420 | 0.001297 |
| [12.50, 36.50)  | 763   | 0.192628  | 566       | 197   | 0.2582     | -0.134523 | 0.003610 | 0.000451 |
| [36.50, 43.50)  | 224   | 0.056551  | 171       | 53    | 0.2366     | -0.018542 | 0.000020 | 0.000002 |
| [43.50, 89.50)  | 1545  | 0.390053  | 1186      | 359   | 0.2324     | 0.005106  | 0.000010 | 0.000001 |
| [89.50, 114.50) | 838   | 0.211563  | 662       | 176   | 0.2100     | 0.134868  | 0.003709 | 0.000463 |
| [114.50, inf)   | 207   | 0.052260  | 181       | 26    | 0.1256     | 0.750487  | 0.023610 | 0.002884 |
| Special         | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
| Missing         | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
|                 | 3961  | 1.000000  | 3037      | 924   | 0.2333     |           | 0.041379 | 0.005099 |

TABLA DE BINNING WOE — VARIABLE «DEUDA\_INGRESO»

| Bin          | Count | Count (%) | Non-event | Event | Event rate | WoE       | IV       | JS       |
|--------------|-------|-----------|-----------|-------|------------|-----------|----------|----------|
| (-inf, 0.08) | 293   | 0.073971  | 255       | 38    | 0.1297     | 0.713764  | 0.030577 | 0.003743 |
| [0.08, 0.16) | 558   | 0.140874  | 453       | 105   | 0.1882     | 0.272018  | 0.009663 | 0.001204 |
| [0.16, 0.37) | 1577  | 0.398132  | 1267      | 310   | 0.1966     | 0.217921  | 0.017802 | 0.002221 |
| [0.37, 0.51) | 654   | 0.165110  | 489       | 165   | 0.2523     | -0.103496 | 0.001817 | 0.000227 |
| [0.51, 0.85) | 674   | 0.170159  | 465       | 209   | 0.3101     | -0.390210 | 0.028516 | 0.003542 |
| [0.85, inf)  | 205   | 0.051755  | 108       | 97    | 0.4732     | -1.082493 | 0.075143 | 0.008960 |
| Special      | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
| Missing      | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
|              | 3961  | 1.000000  | 3037      | 924   | 0.2333     |           | 0.163519 | 0.019897 |

TABLA DE BINNING WOE — VARIABLE «INGRESO\_MENSUAL»

| Bin                    | Count | Count (%) | Non-event | Event | Event rate | WoE       | IV       | JS       |
|------------------------|-------|-----------|-----------|-------|------------|-----------|----------|----------|
| (-inf, 242795.88)      | 210   | 0.053017  | 120       | 90    | 0.4286     | -0.902231 | 0.052230 | 0.006316 |
| [242795.88, 354256.50) | 587   | 0.148195  | 380       | 207   | 0.3526     | -0.582461 | 0.057607 | 0.007101 |
| [354256.50, 464805.09) | 806   | 0.203484  | 579       | 227   | 0.2816     | -0.253561 | 0.013952 | 0.001739 |
| [464805.09, 631639.47) | 917   | 0.231507  | 709       | 208   | 0.2268     | 0.036404  | 0.000304 | 0.000038 |
| [631639.47, 913196.97) | 876   | 0.221156  | 731       | 145   | 0.1655     | 0.427766  | 0.035835 | 0.004445 |
| [913196.97, inf)       | 565   | 0.142641  | 518       | 47    | 0.0832     | 1.209914  | 0.144823 | 0.017074 |
| Special                | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
| Missing                | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
|                        | 3961  | 1.000000  | 3037      | 924   | 0.2333     |           | 0.304750 | 0.036713 |

TABLA DE BINNING WOE — VARIABLE «MORA\_MAX\_12M»

| Bin          | Count | Count (%) | Non-event | Event | Event rate | WoE       | IV       | JS       |
|--------------|-------|-----------|-----------|-------|------------|-----------|----------|----------|
| (-inf, 2.50) | 240   | 0.060591  | 198       | 42    | 0.1750     | 0.360684  | 0.007120 | 0.000885 |
| [2.50, 3.50) | 355   | 0.089624  | 285       | 70    | 0.1972     | 0.214081  | 0.003872 | 0.000483 |
| [3.50, 5.50) | 1173  | 0.296137  | 918       | 255   | 0.2174     | 0.091020  | 0.002394 | 0.000299 |
| [5.50, 7.50) | 1166  | 0.294370  | 878       | 288   | 0.2470     | -0.075227 | 0.001699 | 0.000212 |
| [7.50, 9.50) | 678   | 0.171169  | 504       | 174   | 0.2566     | -0.126392 | 0.002826 | 0.000353 |
| [9.50, inf)  | 349   | 0.088109  | 254       | 95    | 0.2722     | -0.206456 | 0.003960 | 0.000494 |
| Special      | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
| Missing      | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
|              | 3961  | 1.000000  | 3037      | 924   | 0.2333     |           | 0.021870 | 0.002727 |

#### TABLA DE BINNING WOE — VARIABLE «SEGMENTO»

| Bin               | Count | Count (%) | Non-event | Event | Event rate | WoE       | IV       | JS       |
|-------------------|-------|-----------|-----------|-------|------------|-----------|----------|----------|
| ['independiente'] | 1353  | 0.341580  | 1052      | 301   | 0.2225     | 0.061425  | 0.001268 | 0.000158 |
| ['asalariado']    | 1303  | 0.328957  | 1001      | 302   | 0.2318     | 0.008414  | 0.000023 | 0.000003 |
| ['pensionado']    | 1305  | 0.329462  | 984       | 321   | 0.2460     | -0.069729 | 0.001632 | 0.000204 |
| Special           | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
| Missing           | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
|                   | 3961  | 1.000000  | 3037      | 924   | 0.2333     |           | 0.002922 | 0.000365 |

#### TABLA DE BINNING WOE — VARIABLE «UTILIZACION\_LINEA»

| Bin          | Count | Count (%) | Non-event | Event | Event rate | WoE       | IV       | JS       |
|--------------|-------|-----------|-----------|-------|------------|-----------|----------|----------|
| (-inf, 0.10) | 233   | 0.058824  | 200       | 33    | 0.1416     | 0.611896  | 0.018443 | 0.002270 |
| [0.10, 0.25) | 788   | 0.198940  | 637       | 151   | 0.1916     | 0.249576  | 0.011562 | 0.001442 |
| [0.25, 0.45) | 1385  | 0.349659  | 1077      | 308   | 0.2224     | 0.061922  | 0.001318 | 0.000165 |
| [0.45, 0.68) | 1212  | 0.305983  | 892       | 320   | 0.2640     | -0.164768 | 0.008668 | 0.001082 |
| [0.68, inf)  | 343   | 0.086594  | 231       | 112   | 0.3265     | -0.465995 | 0.021040 | 0.002606 |
| Special      | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
| Missing      | 0     | 0.000000  | 0         | 0     | 0.0000     | 0.000000  | 0.000000 | 0.000000 |
|              | 3961  | 1.000000  | 3037      | 924   | 0.2333     |           | 0.061031 | 0.007565 |

#### MATRIZ DE CORRELACIÓN ENTRE VARIABLES

| ingreso_mensual_woe | deuda_ingreso_woe | utilizacion_linea_woe | antiguedad_meses_woe | mora_max_12m_woe |
|---------------------|-------------------|-----------------------|----------------------|------------------|
| 1.000000            | 0.004914          | -0.008224             | 0.014162             | 0.016804         |
| 0.004914            | 1.000000          | 0.009980              | 0.000783             | 0.002489         |
| -0.008224           | 0.009980          | 1.000000              | 0.003123             | 0.019960         |
| 0.014162            | 0.000783          | 0.003123              | 1.000000             | 0.030326         |
| 0.016804            | 0.002489          | 0.019960              | 0.030326             | 1.000000         |

#### ESTABILIDAD DE LAS VARIABLES CANDIDATAS

| feature | woe_column | sample | csi | csi_band | smoothing |
|---------|------------|--------|-----|----------|-----------|
|---------|------------|--------|-----|----------|-----------|

## DETALLE POR OBSERVACIÓN

Esta corrida produjo tablas con una fila por operación que no se reproducen en el informe (son datos, no evidencia de validación) y que **no se exportaron**: `añada.csv` o `xlsx.report.format`s para obtenerlas completas.

- Dataset transformado a WoE — 6.000 filas
- PD calibrada por observación — 6.000 filas
- PD sin calibrar por observación — 6.000 filas
- Puntaje por observación — 6.000 filas
- Dataset WoE de las variables seleccionadas — 6.000 filas

## Anexo C — Parámetros completos

Este anexo reproduce el payload completo de cada etapa: los parámetros efectivos y las métricas estructuradas tal como las publicó cada step, sin resumir.

### C.1 Población, particiones y exclusiones

Artefacto de origen: data.data\_card

#### PARÁMETROS Y PAYLOAD

**bad\_rate** 0.2345

**class\_counts**

```
{
  "bueno": 4593,
  "excluido": 0,
  "indeterminado": 0,
  "malo": 1407
}
```

**data\_hash**

74c5d9ddeca8ff71853623d20ace015d0788564549339d61bfb66f24dfadfcc8

**effective\_config**

```
{
  "load": {
    "backend": "pandas",
    "csv_options": {
      "decimal": ".",
      "encoding": "utf-8",
      "sep": ","
    },
    "file_format": "auto",
    "source": ".nikodym-demo-fixtures-f1/datasets/
consumo_comportamiento.parquet"
  },
  "missing": {
    "max_missing_rate": "0.9900",
    "special_values": []
  },
  "partition": {
    "min_bads_per_partition": 30,
    "strategy": {
      "cohort_col": "cohort",
      "holdout_fraction": "0.2000",
      "oot_cohorts": [
```

```

    "2024Q2"
  ],
  "type": "cohort"
},
"ttdd_includes_excluded": true
},
"schema_": {
  "columns": [
    {
      "coerce": false,
      "dtype": "float",
      "ge": null,
      "isin": null,
      "le": null,
      "name": "ingreso_mensual",
      "nullable": false,
      "required": true,
      "unique": false
    },
    {
      "coerce": false,
      "dtype": "float",
      "ge": null,
      "isin": null,
      "le": null,
      "name": "deuda_ingreso",
      "nullable": false,
      "required": true,
      "unique": false
    },
    {
      "coerce": false,
      "dtype": "float",
      "ge": null,
      "isin": null,
      "le": null,
      "name": "utilizacion_linea",
      "nullable": false,
      "required": true,
      "unique": false
    },
    {
      "coerce": false,
      "dtype": "int",
      "ge": null,
      "isin": null,
      "le": null,

```



```

    "name": "mora_max_12m",
    "nullable": false,
    "required": true,
    "unique": false
  },
  {
    "coerce": false,
    "dtype": "int",
    "ge": null,
    "isin": null,
    "le": null,
    "name": "antiguedad_meses",
    "nullable": false,
    "required": true,
    "unique": false
  },
  {
    "coerce": false,
    "dtype": "str",
    "ge": null,
    "isin": null,
    "le": null,
    "name": "segmento",
    "nullable": false,
    "required": true,
    "unique": false
  },
  {
    "coerce": false,
    "dtype": "str",
    "ge": null,
    "isin": null,
    "le": null,
    "name": "cohorte",
    "nullable": false,
    "required": true,
    "unique": false
  },
  {
    "coerce": false,
    "dtype": "int",
    "ge": null,
    "isin": null,
    "le": null,
    "name": "bad_flag",
    "nullable": false,
    "required": true,

```

```

        "unique": false
      }
    ],
    "index_col": "loan_id",
    "ordered": false,
    "strict": false,
    "unique_keys": null
  },
  "target": {
    "bad_rule": {
      "all_of": [
        {
          "col": "bad_flag",
          "op": "==",
          "value": 1
        }
      ],
      "any_of": []
    },
    "exclusion_rules": [],
    "good_rule": null,
    "indeterminate_rule": null,
    "target_col": "target",
    "window": null
  },
  "type": "standard"
}

```

**exclusions\_by\_reason** {}

**n\_features** 8

**n\_rows** 6000

**partition\_bad\_rates**

```

{
  "desarrollo": "0.2333",
  "fuera_de_modelo": "0.0000",
  "holdout": "0.2367",
  "oot": "0.2371"
}

```

**partition\_sizes**

```

{
  "desarrollo": 3961,
  "fuera_de_modelo": 0,
  "holdout": 1031,
}

```

```
"oot": 1008
}
```

### performance\_window\_months

**source** consumo\_comportamiento.parquet

**target\_col** target

## C.2 Binning WoE y poder predictivo

Artefacto de origen: binning.binning\_card

### PARÁMETROS Y PAYLOAD

#### effective\_config

```
{
  "cat_cutoff": "0.010000",
  "cat_unknown": null,
  "categorical_columns": [
    "segmento"
  ],
  "exclude_columns": [],
  "fail_on_non_binnable": false,
  "feature_columns": [
    "ingreso_mensual",
    "deuda_ingreso",
    "utilizacion_linea",
    "mora_max_12m",
    "antiguedad_meses",
    "segmento"
  ],
  "keep_structural_columns": true,
  "max_n_bins": 6,
  "max_n_prebins": 20,
  "max_pvalue": null,
  "max_pvalue_policy": "consecutive",
  "metric_missing": "empirical",
  "metric_special": "empirical",
  "min_bin_n_event": 1,
  "min_bin_n_nonevent": 1,
  "min_bin_size": "0.050000",
  "min_event_rate_diff": "0.0000",
  "min_n_bins": null,
  "min_prebin_size": "0.050000",
  "mip_solver": "bop",
  "monotonic_trend": "auto_asc_desc",
  "n_jobs": null,
```

```

"output_suffix": "__woe",
"require_optimal": true,
"solver": "mip",
"special_handling": "separate",
"split_digits": null,
"time_limit": 10,
"type": "standard",
"variable_overrides": []
}

```

**iv\_by\_variable**

```

{
  "antiguedad_meses": "0.041379",
  "deuda_ingreso": "0.163519",
  "ingreso_mensual": "0.304750",
  "mora_max_12m": "0.021870",
  "segmento": "0.002922",
  "utilizacion_linea": "0.061031"
}

```

**missing\_handling**

empirical

**monotonicity\_by\_variable**

```

{
  "antiguedad_meses": "descending",
  "deuda_ingreso": "ascending",
  "ingreso_mensual": "descending",
  "mora_max_12m": "ascending",
  "segmento": null,
  "utilizacion_linea": "ascending"
}

```

**n\_variables\_binned**

6

**n\_variables\_requested**

6

**n\_variables\_skipped**

0

**optbinning\_version**

0.20.0

**special\_handling**

separate

**C.3 Selección de variables**Artefacto de origen: `selection.selection_card`**PARÁMETROS Y PAYLOAD****dependency\_versions**

```
{
  "numpy": "2.4.6",
  "pandas": "2.3.3",
  "scikit-learn": "1.7.2",
  "scipy": "1.18.0",
  "statsmodels": "0.14.6"
}
```

## effective\_config

```
{
  "compute_univariate_metrics": true,
  "correlation": {
    "clustering_method": "none",
    "enabled": true,
    "method": "pearson",
    "threshold": "0.750000"
  },
  "exclude_columns": [],
  "fail_if_no_features": true,
  "feature_columns": "*",
  "force_exclude": [],
  "force_include": [],
  "keep_structural_columns": true,
  "max_iv": "0.500000",
  "max_iv_action": "flag",
  "min_auc": null,
  "min_gini": null,
  "min_iv": "0.020000",
  "min_ks": null,
  "priority_order": [
    "iv",
    "auc",
    "ks",
    "name"
  ],
  "stability": {
    "action": "report_only",
    "enabled": false,
    "review_threshold": "0.250000",
    "smoothing": "0.000001",
    "stable_threshold": "0.100000"
  },
  "type": "standard",
  "vif": {
    "add_intercept": true,
    "enabled": true,
    "max_iterations": null,
    "threshold": "5.000000"
  }
}
```

```
}
}
```

**excluded\_by\_reason**

```
{
  "low_iv": 1
}
```

**high\_iv\_flags**

```
[]
```

**max\_abs\_correlation\_after\_selection**

0.030326

**max\_vif\_after\_selection**

1.001595

**n\_candidates**

6

**n\_excluded**

1

**n\_selected**

5

**selected\_features**

```
[
  "antiguedad_meses",
  "deuda_ingreso",
  "ingreso_mensual",
  "mora_max_12m",
  "utilizacion_linea"
]
```

**stability\_flags**

```
[]
```

**thresholds**

```
{
  "correlation.clustering_method": "none",
  "correlation.method": "pearson",
  "correlation.threshold": "0.750000",
  "max_iv": "0.500000",
  "max_iv_action": "flag",
  "min_auc": null,
  "min_gini": null,
  "min_iv": "0.020000",
  "min_ks": null,
  "stability.action": "report_only",
  "stability.review_threshold": null,
  "stability.stable_threshold": null,
  "vif.threshold": "5.000000"
}
```

## C.4 Modelo PD

Artefacto de origen: `model.model_card`

### PARÁMETROS Y PAYLOAD

#### `dependency_versions`

```
{
  "numpy": "2.4.6",
  "pandas": "2.3.3",
  "scikit-learn": "1.7.2",
  "scipy": "1.18.0",
  "statsmodels": "0.14.6"
}
```

#### `effective_config`

```
{
  "alpha": "0.050000",
  "engine": "logit",
  "fail_if_no_features": true,
  "fit_intercept": true,
  "fit_maxiter": 100,
  "force_exclude": [],
  "force_include": [],
  "iv_contribution": {
    "action": "flag",
    "threshold": "0.900000"
  },
  "optimizer": "newton",
  "sign_policy": {
    "action": "flag",
    "expected_beta_sign": "negative",
    "fail_on_forced_inverted": false
  },
  "stepwise": {
    "criterion": "wald_pvalue",
    "direction": "bidirectional",
    "enabled": true,
    "entry_p_value": "0.050000",
    "exit_p_value": "0.050000",
    "max_iter": 100,
    "min_features": 1
  },
  "tol": "0.000000",
  "type": "standard"
}
```

#### `engine`

logit

**final\_features**

```
[
  "antiguedad_meses",
  "deuda_ingreso",
  "ingreso_mensual",
  "mora_max_12m",
  "utilizacion_linea"
]
```

**fit\_statistics**

```
{
  "aic": "3898.715734",
  "bic": "3936.421245",
  "converged": true,
  "llr": "416.536024",
  "llr_p_value": "0.000000",
  "log_likelihood": "-1943.357867",
  "n_events_dev": 924,
  "n_iterations": 6,
  "n_nonevents_dev": 3037,
  "n_obs_dev": 3961,
  "null_log_likelihood": "-2151.625879",
  "optimizer": "newton",
  "pseudo_r2_mcfadden": "0.096796"
}
```

**iv\_contribution\_flags**

```
[
  "ingreso_mensual"
]
```

**n\_candidates**

5

**n\_final\_features**

5

**sign\_flags**

[]

**thresholds**

```
{
  "alpha": "0.050000",
  "engine": "logit",
  "entry_p_value": "0.050000",
  "exit_p_value": "0.050000",
  "fail_if_no_features": "true",
  "fit_intercept": "true",
  "fit_maxiter": "100.000000",
  "iv_contribution.action": "flag",
  "iv_contribution.threshold": "0.900000",
  "optimizer": "newton",
  "sign_policy.action": "flag",
}
```



```
"sign_policy.expected_beta_sign": "negative",
"stepwise.criterion": "wald_pvalue",
"stepwise.direction": "bidirectional",
"stepwise.max_iter": "100.000000",
"stepwise.min_features": "1.000000"
}
```

## C.5 Scorecard

Artefacto de origen: scorecard.card

### PARÁMETROS Y PAYLOAD

#### dependency\_versions

```
{
  "numpy": "2.4.6",
  "pandas": "2.3.3"
}
```

#### effective\_config

```
{
  "clip": false,
  "intercept_allocation": "uniform",
  "max_score": null,
  "min_score": null,
  "output_suffix": "__points",
  "pdo": "20.000000",
  "point_overrides": [],
  "rounding_method": "nearest_integer",
  "score_column": "score",
  "score_direction": "higher_is_lower_risk",
  "target_odds": "50.000000",
  "target_score": "600.000000",
  "type": "standard"
}
```

|                        |            |
|------------------------|------------|
| <b>factor</b>          | 28.853901  |
| <b>max_score</b>       | —          |
| <b>min_score</b>       | —          |
| <b>n_variables</b>     | 5          |
| <b>offset</b>          | 487.122876 |
| <b>overrides_count</b> | 0          |
| <b>pdo</b>             | 20.000000  |

**points\_columns**

```
[
  "antiguedad_meses__points",
  "deuda_ingreso__points",
  "ingreso_mensual__points",
  "mora_max_12m__points",
  "utilizacion_linea__points"
]
```

**rounding\_method**

nearest\_integer

**score\_column**

score

**score\_direction**

higher\_is\_lower\_risk

**target\_odds**

50.000000

**target\_score**

600.000000

## C.6 Calibración

Artefacto de origen: calibration.card

### PARÁMETROS Y PAYLOAD

**anchor\_kind**

through\_the\_cycle

**anchor\_source**

business\_input

**calibrated\_mean\_pd\_dev**

0.2000

**dependency\_versions**

```
{
  "numpy": "2.4.6",
  "pandas": "2.3.3",
  "scipy": "1.18.0"
}
```

**effective\_config**

```
{
  "anchor_kind": "through_the_cycle",
  "anchor_source": "business_input",
  "fit_partition": "desarrollo",
  "linear_predictor_calibrated_column":
"linear_predictor_calibrated",
  "linear_predictor_column": "linear_predictor",
  "max_abs_offset": null,
  "max_iter": 100,
  "method": "intercept_offset",
  "min_fit_rows": 30,
  "partition_column": "partition",
```

```

"pd_calibrated_column": "pd_calibrated",
"pd_raw_column": "pd_raw",
"require_both_classes_for_supervised": true,
"target_column": "target",
"target_pd": "0.200000",
"target_tolerance": "0.000000",
"type": "standard"
}

```

|                                  |                  |
|----------------------------------|------------------|
| <b>fit_partition</b>             | desarrollo       |
| <b>intercept</b>                 | —                |
| <b>method</b>                    | intercept_offset |
| <b>n_fit</b>                     | 3961             |
| <b>observed_default_rate_dev</b> | 0.2333           |
| <b>offset</b>                    | -0.218447        |
| <b>pd_calibrated_column</b>      | pd_calibrated    |
| <b>pd_raw_column</b>             | pd_raw           |
| <b>ranking_preserved</b>         | true             |
| <b>raw_mean_pd_dev</b>           | 0.233274         |
| <b>slope</b>                     | —                |
| <b>target_pd</b>                 | 0.200000         |
| <b>ties_created</b>              | 0                |

## C.7 Desempeño y discriminación

Artefacto de origen: `performance.card`

### PARÁMETROS Y PAYLOAD

**bands\_by\_partition**

```

{
  "desarrollo": "ok",
  "holdout": "ok",
  "oot": "ok"
}

```

**dependency\_versions**

```

{
  "numpy": "2.4.6",
  "pandas": "2.3.3",
}

```

```
"scikit-learn": "1.7.2"
}
```

**effective\_config**

```
{
  "evaluation_source": "pd_calibrated",
  "min_events_per_partition": 1,
  "min_rows_per_partition": 30,
  "n_deciles": 10,
  "optional_thresholds": {},
  "partition_column": "partition",
  "partitions": [
    "desarrollo",
    "holdout",
    "oot"
  ],
  "pd_column": "pd_calibrated",
  "schema_version": "1.0.0",
  "score_column": "score",
  "score_direction": "higher_is_lower_risk",
  "target_column": "target",
  "type": "standard"
}
```

**evaluation\_source**

pd\_calibrated

**max\_metrics\_by\_partition**

```
{
  "desarrollo": {
    "auc": "0.712346",
    "gini": "0.424692",
    "ks": "0.320144"
  },
  "holdout": {
    "auc": "0.694646",
    "gini": "0.389292",
    "ks": "0.311892"
  },
  "oot": {
    "auc": "0.656096",
    "gini": "0.312192",
    "ks": "0.251906"
  }
}
```

**n\_deciles**

10

**partitions**

```
[
  "desarrollo",
  "holdout",
  "oot"
]
```

**score\_direction**

higher\_is\_lower\_risk

**thresholds**

{}

**MÉTRICAS ESTRUCTURADAS****discrimination**

```
{
  "effective_deciles_by_partition": {
    "desarrollo": 10,
    "holdout": 10,
    "oot": 10
  },
  "not_evaluable_reasons_by_partition": {},
  "threshold_flags_by_partition": {}
}
```

**C.8 Estabilidad**Artefacto de origen: `stability.card`**PARÁMETROS Y PAYLOAD****bands\_by\_comparison**

```
{
  "dev_vs_holdout": "stable",
  "dev_vs_oot": "stable"
}
```

**comparisons**

```
[
  "dev_vs_holdout",
  "dev_vs_oot"
]
```

**csi\_source**

score\_points

**dependency\_versions**

```
{
  "numpy": "2.4.6",
  "pandas": "2.3.3",
  "pandera": "0.32.0"
}
```

**effective\_config**

```
{
  "comparisons": [
    "dev_vs_holdout",
    "dev_vs_oob"
  ],
  "csi_bins": 10,
  "csi_source": "score_points",
  "include_pd_stability": true,
  "partition_column": "partition",
  "pd_column": "pd_calibrated",
  "psi_bins": 10,
  "psi_review_threshold": "0.250000",
  "psi_stable_threshold": "0.100000",
  "schema_version": "1.0.0",
  "score_column": "score",
  "score_direction": "higher_is_lower_risk",
  "smoothing": "0.000001",
  "temporal_axis": "period",
  "temporal_column": null,
  "temporal_freq": "M",
  "type": "standard"
}
```

**max\_psi\_by\_comparison**

```
{
  "dev_vs_holdout": "0.013182",
  "dev_vs_oob": "0.006792"
}
```

**psi\_bins**

10

**review\_threshold**

0.250000

**score\_direction**

higher\_is\_lower\_risk

**stable\_threshold**

0.100000

**worst\_csi\_feature**

mora\_max\_12m\_\_points

**worst\_csi\_value**

0.010209

**MÉTRICAS ESTRUCTURADAS****stability**

```
{
  "csi_features": [
    "antiguedad_meses__points",
    "deuda_ingreso__points",
    "ingreso_mensual__points",
    "mora_max_12m__points",
  ]
}
```

```

    "utilizacion_linea__points"
  ],
  "include_pd_stability": true,
  "n_periods": 6,
  "temporal_axis": "period"
}

```

## C.9 Validación formal

Artefacto de origen: validation.card

### PARÁMETROS Y PAYLOAD

#### dependency\_versions

```

{
  "numpy": "2.4.6",
  "pandas": "2.3.3",
  "scipy": "1.18.0"
}

```

#### effective\_config

```

{
  "backtesting": {
    "alpha": "0.050000",
    "enabled": false,
    "one_sided": true,
    "parameters": [
      "pd",
      "lgd",
      "ead"
    ],
    "pd_test": "jeffreys",
    "realised_ead_col": "realised_ead",
    "realised_lgd_col": "realised_lgd",
    "realised_pd_col": "realised_default",
    "segment_col": "portfolio"
  },
  "calibration": {
    "alpha": "0.050000",
    "binomial_by_grade": false,
    "brier": true,
    "grade_col": "grade",
    "hl_grouping": "deciles",
    "hl_n_groups": 10,
    "hosmer_lemeshow": true,
    "min_rows_per_group": 30,
    "partition_column": "partition",

```

```

    "pd_column": "pd_calibrated",
    "pd_test": "jeffreys",
    "target_column": "target",
    "traffic_light_green_alpha": "0.050000",
    "traffic_light_red_alpha": "0.010000"
  },
  "discrimination": {
    "consume_performance": true,
    "partitions": [
      "desarrollo",
      "holdout",
      "oot"
    ]
  },
  "fail_on_falta_dato": true,
  "families": [
    "discrimination",
    "calibration",
    "stability"
  ],
  "schema_version": "1.0.0",
  "stability": {
    "consume_stability": true,
    "psi_review_threshold": "0.250000",
    "psi_stable_threshold": "0.100000"
  },
  "type": "standard"
}

```

falta\_dato

[]

families\_run

```

[
  "discrimination",
  "calibration",
  "stability"
]

```

model\_ref

preset-estandar-consumo

n\_failed

3

n\_tests

3

overall\_status

fail

**MÉTRICAS ESTRUCTURADAS**

validation



```
{
  "families_run": [
    "discrimination",
    "calibration",
    "stability"
  ],
  "n_failed": 3,
  "n_tests": 3,
  "not_evaluable_grades": [],
  "overall_status": "fail",
  "traffic_light": {
    "amber": 0,
    "green": 0,
    "red": 0
  }
}
```